

FINAL ENGINEERING DESIGN REPORT

Fixed Wing Team 5



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Nomenclature

ACN	Aircraft classification number
APU	Auxiliary power unit
AR	Aspect ratio
BFL	Balance field length
CFRP	Carbon fibre reinforced polymer
CG	Centre of gravity
COC	Cash operating cost
DOC	Direct operating cost
EASA	European Aviation Safety Agency
ECS	Environmental control system
EHA	Electro hydrostatic actuators
ELAC	Elevator aileron computer
FAA	Federal Aviation Authority
FAC	Flight augmentation computer
GA	General arrangement
HTP	Horizontal tailplane
JAR	Joint aviation requirements
MEA	More electric aircraft
MLG	Main landing gear
MLW	Maximum landing weight
MRW	Maximum ramp weight
MSP	Manufacturer's study price
MTOW	Maximum take-off weight
MWE	Manufacturer's weight empty
MZFW	Maximum zero fuel weight
NLG	Nose landing gear
NRC	Non-recurring costs
OWE	Operational weight empty
RAT	Ram air turbine
RC	Recurring costs
SAF	Sustainable aviation fuel
SEC	Spoiler elevator computer
TR	Taper ratio
TRU	Transformer rectifier unit
VFG	Variable frequency generator
VFSG	Variable frequency starter generator
VLM	Vortex lattice method
VTP	Vertical tailplane

Symbols

C_{D0}	Profile drag coefficient
C_L	Lift coefficient
$C_{L_{max}}$	Maximum lift coefficient
$C_{L_{to}}$	Take-off lift coefficient
V_a	Approach speed
V_c	Design cruise speed
V_d	Design dive speed
V_s	Stall speed
λ	Dynamic reaction factor
μ	Take off friction coefficient

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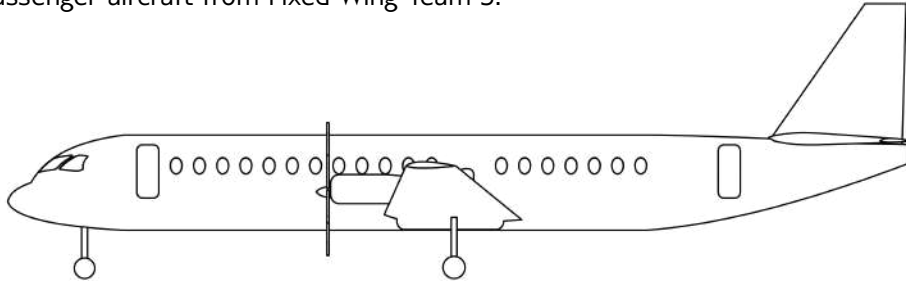
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CONFIGURATION DOCUMENT

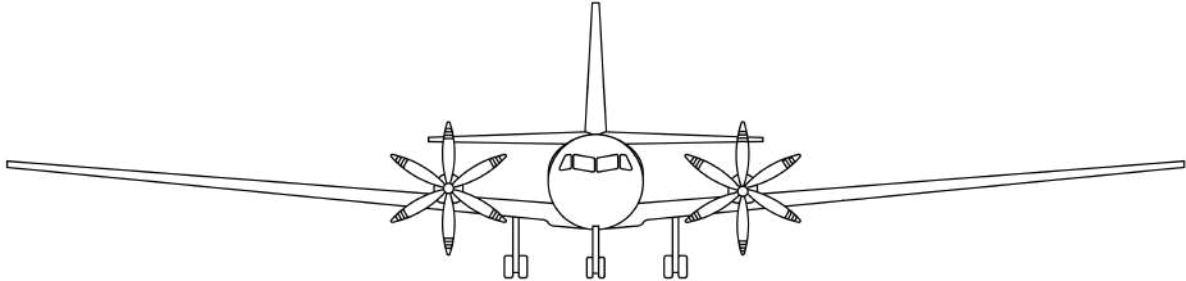
1 EXECUTIVE SUMMARY AND WALK AROUND CHART

This report details the conception, development and final detailed design of Falco, the next-generation regional passenger aircraft from Fixed Wing Team 5.



Falco aims to service a gap in the current airliner market; carrying 120 PAX over a range of 800 nm, it sits between present regional turboprops and short-haul turbofans. In essence, it has the passenger capability of an Airbus A220 or Embraer E190, but is optimised for much shorter flights, currently serviced by the ATR72 and ageing Dash8-Q400.

The design philosophy centres around minimising its environmental impact, focusing on reducing fuel burn, emissions, noise and contrail formation. Intended for low-cost intra-European airlines, it also aims to maximise passenger and airline acceptability, primarily by keeping DOC low and airport compatibility high. Decisions along the development path detailed in this report were made in accordance with these design drivers. For example, the choice of turboprop engines provided a 13% block fuel saving, whilst the cruise altitude of 25,000 ft was chosen to mitigate contrail formation.



When benchmarked against the ATR72-600 on a 300 nm mission, Falco provides a 16% saving in fuel burn per seat, a 7% saving in CO₂, and a 17% saving in DOC, cementing it as a true modern competitor in this market. The aircraft meets the strict Heathrow noise quotas and has a quiet cabin at below 80 dB. Falco can be purchased with the option of built-in stairs, reducing turnaround time, as well as folding wingtips, allowing it to fit into ICAO Code C.

Analysis of the current market dictates that Falco can capture 40% of the demand for new turboprops over the next 20 years, translating to 1000 units at an MSP of \$30M. Its break-even number is 838, demonstrating that large profits of around \$4.9B can be made from Falco.



120 PAX

Range: 800 nmi

Mach. Cruise: 0.47

Alt. Cruise: 25,000 ft

MTOW: 39,700 kg

Vs. ATR 72 (300 nmi)

- Fuel burn: -16%
- CO₂: -7%
- DOC: -17%

Aerodynamics:

- Span = 40.5 m
- AR = 12.25
- L/D (max) = 20.3

Propulsion:

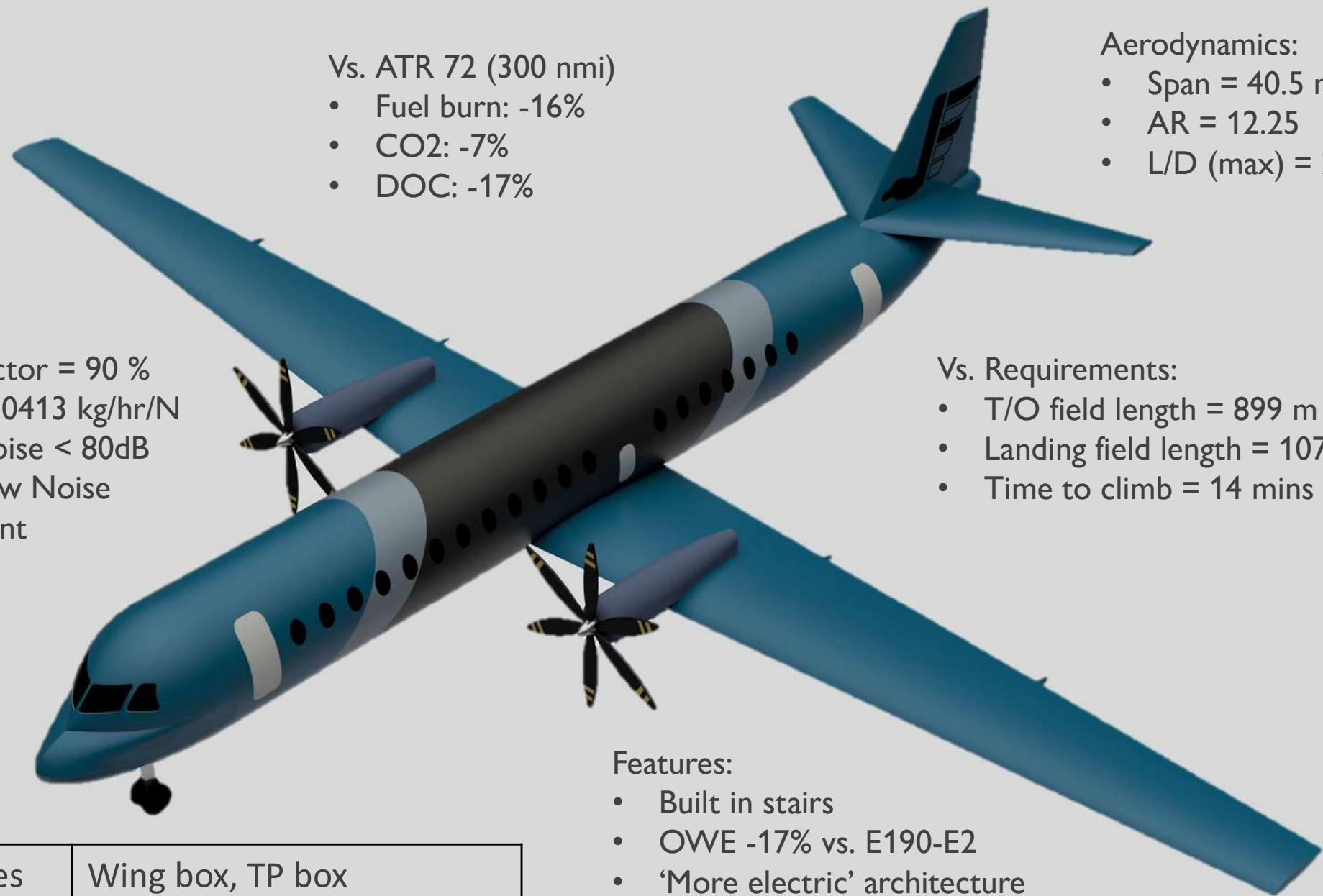
- Scale Factor = 90 %
- SFC = 0.0413 kg/hr/N
- Cabin noise < 80dB
- Heathrow Noise Compliant

Vs. Requirements:

- T/O field length = 899 m
- Landing field length = 1074 m
- Time to climb = 14 mins

Features:

- Built in stairs
- OWE -17% vs. E190-E2
- 'More electric' architecture
- LIDAR gust alleviation
- Fly-by-wire
- Folding wingtips to fit 36m ICAO Type C



Composites

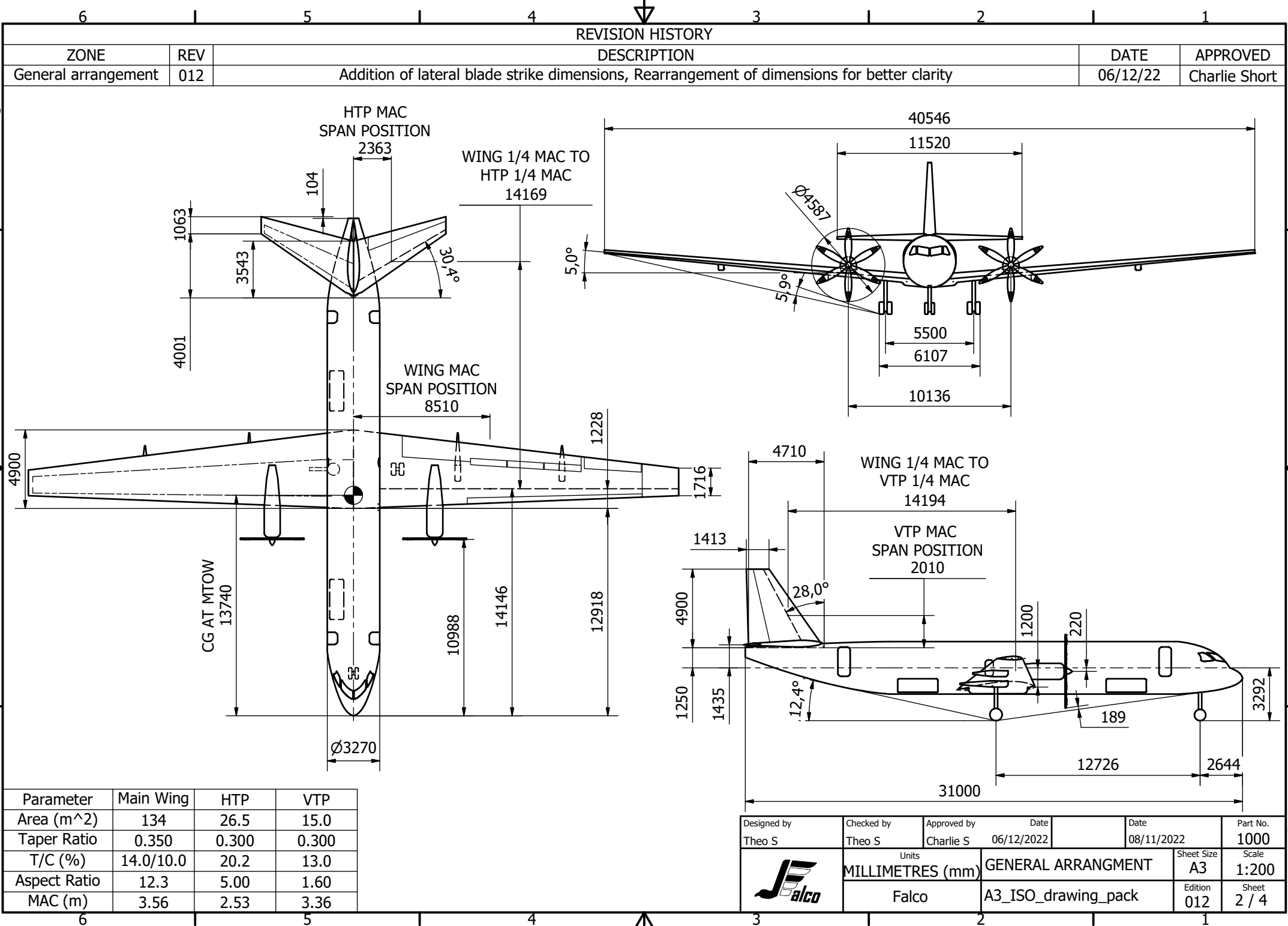
Wing box, TP box

Aluminium

Wing skin, Fuselage

Titanium

LG & Engines



2 AIRCRAFT DATA SHEET

Item	Value	Units	Comments
Configuration	Low Wing, Turboprop, Conventional Tail		
Maximum Take-off Mass	39700	kg	MRW = 39,725
Operators Mass Empty	24075	kg	
Fuel for design mission 800 nm (including reserves)	4250	kg	
Fuel for 300 nm (excluding reserves)	1170	kg	
Payload for Design Mission	11400	kg	
Maximum Landing Mass	37960	kg	
Maximum Zero Fuel Mass	35475	kg	
Maximum Fuel Capacity	4250	kg	
Cruise Mach no.	0.47		
VMO / MMO	357 / 0.54	kts CAS	
Initial Cruise Altitude	25000	ft	
L/D (at start of Cruise for 500 nm mission)	16.1		
Take-Off Field Length	899	m	Balanced = 1430 m
Landing Field Length	1074	m	
Approach Speed	110	kts	
Fuselage Length	31.0	m	
Fuselage Width (or diameter if circular)	3.27	m	
Fuselage Height or Depth	3.27	m	
Wing Area	134.2	m ²	
Span	40.5	m	
Aspect Ratio	12.25		
Thickness of Wing @ outer wing (t/c)	0.10		
Taper Ratio	0.35		
Mean Aerodynamic Chord	3.57	m	
Sweep @ 1/4 Chord	0	deg	
High Lift System (Trailing Edge)	Single Slotted Flap		
High Lift System (Leading Edge)	Slat		
$C_{L_{max}}$ Landing	2.54		
Distance of wing LE at root from nose	12.9	m	
Distance of wing MAC from datum	8.51	m	Datum: a/c centreline
Tailplane Area	26.5	m ²	
Span	11.5	m	
Aspect Ratio	5.0		
1/4 Chord MAC (wing) to 1/4 Chord MAC (tailplane)	14.2	m	
Fin Area	15.0	m ²	
Span	4.90	m	
Aspect Ratio	1.6		
1/4 Chord MAC (wing) to 1/4 Chord MAC (fin)	14.2	m	
Number of Engines	2		
Engine position	Wing Mounted		
Engine Shaft Power (per Engine) SLS	4360	kW	
Thrust (per Engine) SLS	12700	lb	
Engine Thrust Scale Factor	0.9		
Propeller Diameter	4.59	m	
SFC at mid cruise 300 nm mission	0.0413	kg/N.hr	
Main Landing Gear Tyre Size	27.75 x 8.75	inches	
Main Landing Gear Geometry for ACN	5500	mm	
ACN	25.1		
Height of Fuselage Datum above Ground	1.66	m	Ground clearance
CG range at MTOW	6 - 36	% MAC	
Aircraft NRC	1400	\$M	
Aircraft RC	23000	\$M	
Manufacture's Study Price	30.0	\$M	With 20% profit
Engine Price (per engine)	1.9	\$M	
Aircraft DOC	5180	\$/trip	300 nm, 120PAX
Turnaround time	25	mins	
Refuel time	3.5	mins	at 1000 L/min
Trim Tank yes/no	no		

FORE

AFT

ENLARGED TYPE C

508

TYPE III

ENLARGED TYPE C

CS
TOCS
TO

ENLARGED TYPE C

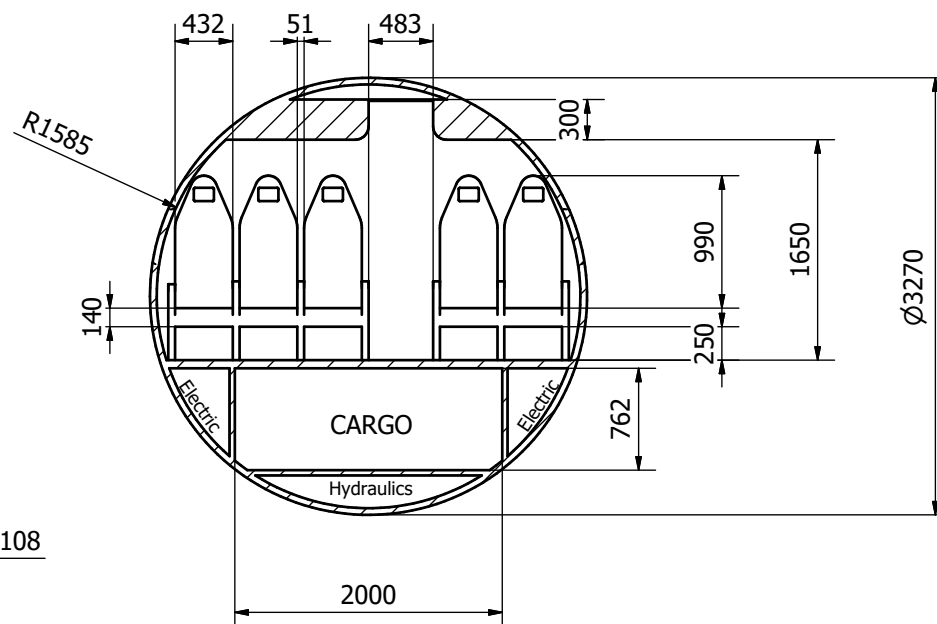
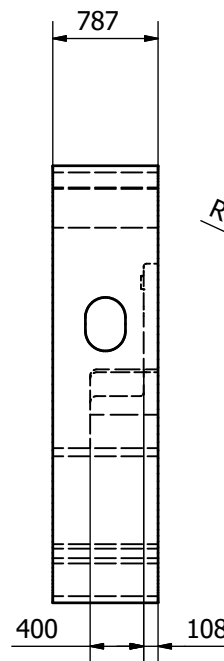
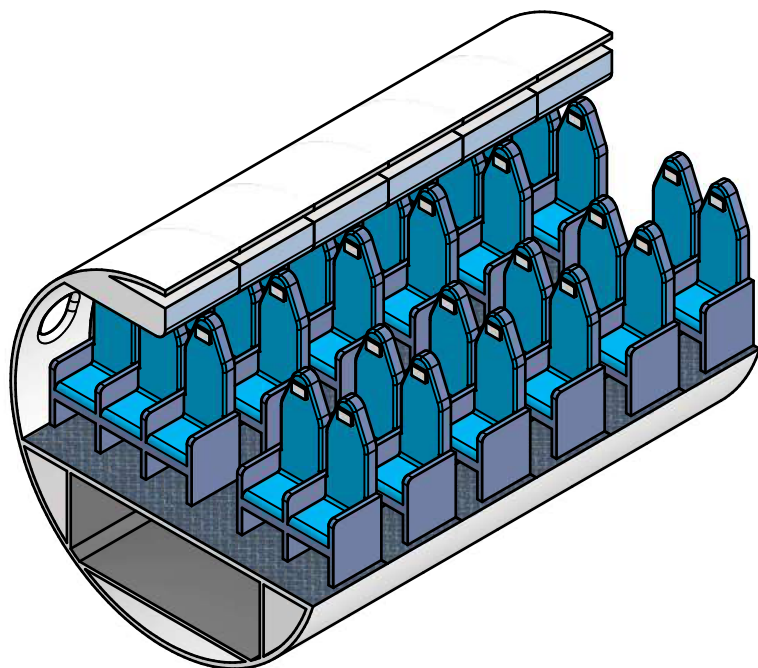
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TYPE III

ENLARGED TYPE C

Item	Acronym
Half Trolley	HT
Crew Seat	CS
Toilet	TO

1:100 SCALE FLOORPLAN



REVISION HISTORY

ZONE	REV	DESCRIPTION	DATE	APPROVED
Cabbin Arrangement	008	Visual update to floorplan, addition of isometric view	04/12/22	Charlie Short

Designed by	Checked by	Approved by	Date	Date	Part No.
Theo S	Theo S	Charlie S	04/12/2022	08/11/2022	1100
	Units	CABIN ARRANGEMENT			Sheet Size
	MILLIMETRES (mm)				A3
	Falco	A3_ISO_drawing_pack	Edition	Sheet	
		008	3 / 4		

3 MARKET ANALYSIS AND COMPETITIVE POSITION

The first step in the market analysis was to identify similar competitor aircraft, however no current aircraft match the given specification for both range and passenger numbers. Figure 1 shows the design point, marked 'AVDASI4', as a clear outlier. The ATR72, Dash8 Q400, A220-100 and E190-E2 are the current market leaders in short-range regional flight, therefore these were chosen as comparable aircraft. The ATR72 and Dash8 Q400 are similar in range whilst the A220-100 and E190-E2 have comparable passenger loads.

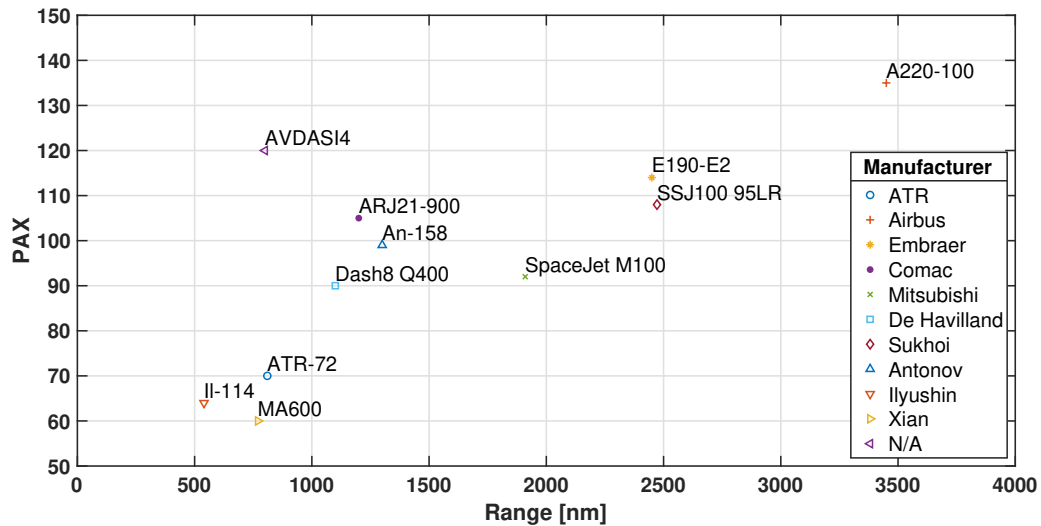


Figure 1: Range vs. Passenger Payload of Competitor Regional and Short-Haul Aircraft

Figure 3a shows the top 25 most popular intra-European flight distances [1]; most flights are between 0-400 nm, with all of them under 1000 nm. These routes are typically flown by aircraft optimised for much longer ranges [2]. By 2028 the volume of flights in Europe is expected to be up to 18% higher than pre-COVID levels [3]. With the ever-present need to reduce environmental impact across all sectors, Falco should be ideal for servicing this increased demand with greater fuel efficiency when it enters service in 2030. The payload range diagram shown in Figure 2 again highlights how Falco is optimised for shorter-range regional flights but with a higher payload capacity than what is currently available.

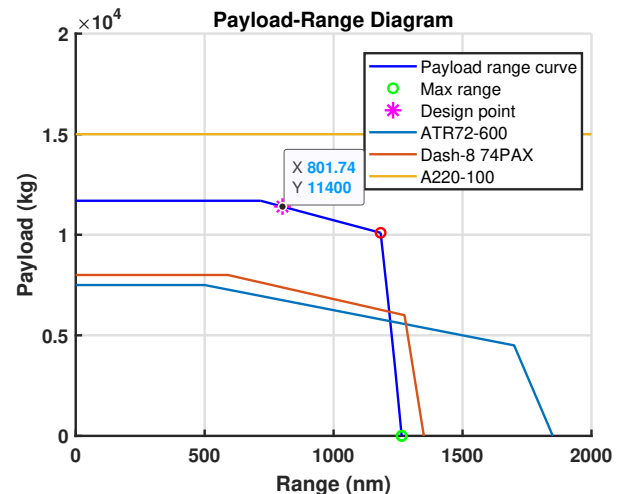
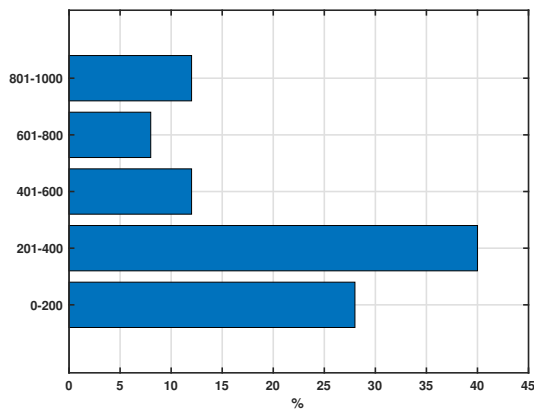
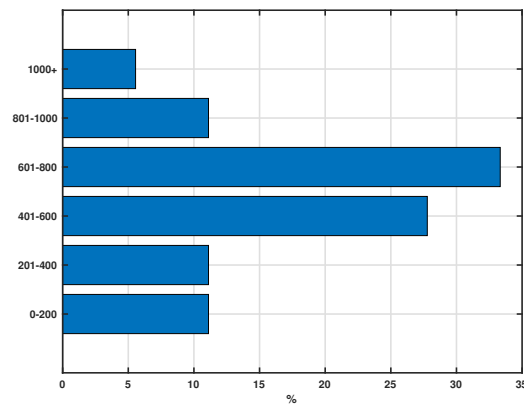


Figure 2: Falco Payload Range Diagram

Although the specification is to design for the European customer, following further research into our position in the market we realised there is great benefit in considering Asia as well. ATR forecasts that there is demand for 2450 new turbo-props over the next 20 years, of which 51% will enter the fast-growing Asian market [4]. Furthermore, research into the busiest routes found that of the top 20 globally, 18 were domestic Asian flights [5], with similar route distance suitability to that of Europe as shown in Figure 3b. This suggests Falco could capture a large portion of the Asian market.



(a) Top 25 European Routes



(b) Top 18 Asian Routes

Figure 3: Distances of the Most Popular Flights in Their Relevant Markets [nm]

4 CONCEPT SELECTION AND JUSTIFICATION

4.1 Requirements Analysis and Falco's Design Drivers

From careful review and discussion of the specification, a design philosophy was set out. At the core of this was the design ethos, minimising environmental impact whilst maximising airline and passenger acceptability. This was extended to outline the key design drivers, presented in Figure 4, to be referred to throughout the initial and more detailed design phases. To help meet these drivers, quantified project targets were discussed and evaluated through the consideration of historic trends between past generations of similar aircraft. It was decided that minimising fuel burn was the most important driver and hence Falco aims to beat competitor aircraft on fuel burn and emissions by 20%. Overall aircraft purchase price is important in the regional market as many airlines run on budget business models, with tight profit margins. Many low-cost and regional airlines fly multiple flights a day and the amount of time an aircraft spends on the ground between flights directly affects their profits [6]. Therefore, keeping turnaround time as low as possible is important so a target of under 30 minutes was set.

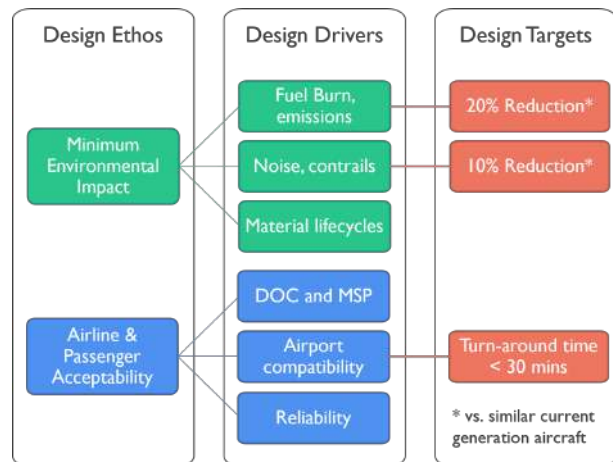


Figure 4

4.2 Concepts

In its most simplistic form, overall aircraft efficiency is a function of three things; aircraft mass, propulsive efficiency and aerodynamics. With the engine parameters pre-selected for this project, our conceptual design phase initially aimed to develop aircraft with the lowest mass, and best possible aerodynamic efficiency. The design drivers were then factored in to assess each concept in the context of real-world operation, and ultimately move forward with the best overall design.

Due to the significant impact on the entire aircraft configuration, the first decision considered was

turbofan vs turboprop. Using a performance analysis tool with an arbitrary airframe it was found that a turboprop would have a 13% lower fuel burn than a turbofan for our 300 nm mission (Figure 5). Turboprops were used for all but one of the concepts, which was included as a reference.

The first concept considered was a conventional turbofan. Whilst turbofans had been ruled out early in the design process, this concept was still included to represent a competitor aircraft – the Airbus A220.

A wide range of aerodynamic solutions were evaluated before settling on the final design. Initially, radical concepts such as a flying wing were considered, due to the potential for improved lift-to-drag performance and reduced fuel burn. However, when assessing this concept in the context of real-world operation, it was quickly realised that the design would run into issues regarding airport compatibility, cabin pressurisation and high project development costs. This meant the flying wing concept would not comply with the design drivers.

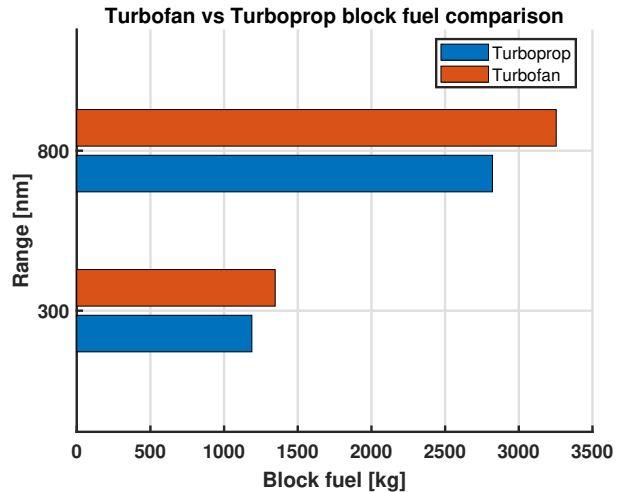


Figure 5: Fuel Burn Turboprop vs. Turbofan

A high-wing T-tail design seemed a strong option, providing adequate propeller ground clearance, however, ground stability requirements were expected to cause issues with landing gear integration. Preliminary calculations also predicted a substantial aerodynamic penalty from the blister required to house the wide landing gear.

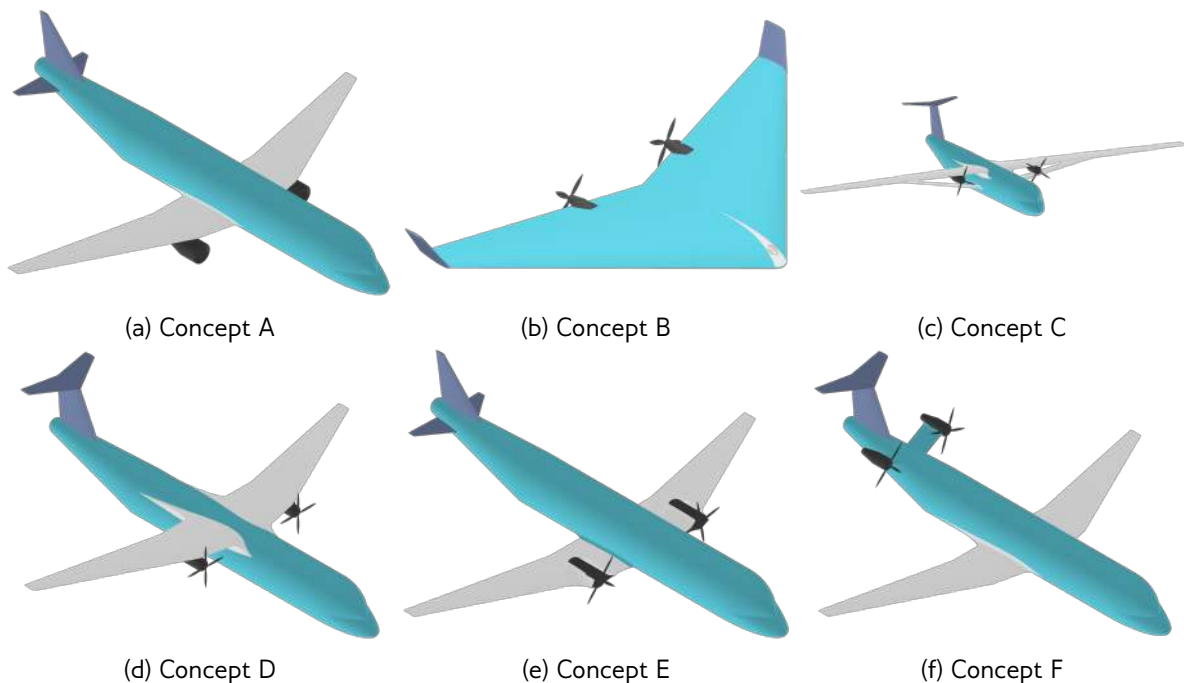


Figure 6: The Six Concepts Derived From Our Research

A high aspect ratio strut supported wing design was also evaluated, however any aerodynamic benefits from the reduced induced drag were expected to be negated by an increase in overall mass and small fuel savings on such a short-range mission. Both high wing concepts necessitated the use of a t-tail

to ensure the horizontal tailplane would be clear from any turbulent airflow from the propellers and the wing downwash. In addition to adding mass to the empennage assembly it also posed the safety risk of a deep stall condition.

Similar to Embraer's 'Next-Generation Turboprop' concept [7], rear engine positioning was investigated. Whilst it posed advantages such as shortened landing gear and the elimination of any wing and engine interference issues, upon closer inspection this design suffers from a range of issues that reduce the overall efficiency of the aircraft. Firstly, the rear mounted engines require significant strengthening of the empennage assembly and incorporate heavy pylons to sufficiently support the load. The high proportion of mass at the rear of the aircraft contributes to a considerably aft CG position. This in turn requires an aft mounted wing, reducing the moment arm and thus increasing the size of the tail surfaces. From the impact of the reduction in moment arm alone it was estimated that the tail masses would be up to 10% higher than a comparable aircraft with wing mounted engines. Mounting the engines at the rear also lacks the structural benefit of wing mounted engines, which provide a reduction in wing root bending moment. Overall, this concept was predicted to yield a much higher structural mass than the alternatives; when considering the importance of low mass on the design, this configuration was decided against.

To assist with the overall concept selection decision, a weighted ranking matrix was created with the requirements and design drivers in mind to determine the most suitable concept. Each parameter was weighted 1-3 depending on its importance in achieving the project design drivers. For example, environmental impact and cruise performance were both given weightings of 3, whereas family development was deemed less significant and only given a weighting of 1. Each concept was given a rating for each parameter using a 1-3-9 scale, used instead of a traditional 1-3 to better differentiate the optimal concepts from those that only moderately fulfil the given parameter.

New Concept Rating		A	B	C	D	E	F
Requirements	Weight						
Manufacturability	2	9	1	1	3	3	1
Ease of maintenance	2	3	9	1	3	3	3
Takeoff/landing performance	3	9	9	3	9	9	9
Cruise performance	3	3	9	9	3	3	9
Airport handling	3	3	1	3	9	3	3
Stability and control performance	2	9	1	3	3	9	3
Weight	3	3	3	1	1	9	3
Safety/reliability	3	9	1	3	9	9	9
Noise	1	3	9	3	1	3	3
Environmental Impact	3	3	9	9	9	9	9
R&D costs and timeline	1	9	1	1	3	3	3
Family development	1	9	1	3	9	9	3
Unweighted Total		72	54	40	62	72	58
Weighted Total		162	138	102	151	171	158

Figure 7: Concept Down Selection Matrix

To assess the impact of inaccurate weightings being assigned, sensitivity studies were carried out on the matrix table using a script that varied the weightings by +/- 20%. This resulted in concept E being chosen as the most suitable concept.

The chosen concept has several advantages. The conventional tail and engine placement ensure that structural mass is kept to a minimum, whilst providing safety benefits such as a significantly reduced risk of deep stall (relative to a T-tail). Turboprop engines provide improved fuel burn relative to equivalent turbofans. The conventional nature of this concept keeps development costs low, and ensures airport and passenger acceptability. The low wing allows wider landing gear placement than for a high wing concept, improving ground stability.

5 DESCRIPTION OF FINAL DESIGN

5.1 An Overview

5.1.1 Cabin Sizing

To begin the design of Falco, the cabin and fuselage were sized following analytical methods proposed by Nita and Scholz [8]. Initially, a trade-off study linking the number of seats abreast with changes in fuselage mass was considered, before extending the analysis to changes in wetted area and hence profile drag. This study favoured 4, 5 and 6 abreast configurations, however, there was insufficient data to make a decision. Therefore, a comparison of fuel burn and DOC was carried out, aligning with our design drivers. This concluded that a 5-abreast cabin was optimal, giving a close to identical fuel burn with a 1.25% saving in DOC over a 300 nm mission, as seen in Figure 8. The seat pitch, galleys, toilets and number of crew were all sized to reduce the floor space required, aiming at keeping mass as low as possible. Falco uses a total of 4 oversized type-C and 2 type-III (overwing) emergency exits, giving it a passenger exit rating of 165. This leaves sufficient room for up to 9 additional rows, without changing the doors, if a stretch variant was to be desired.

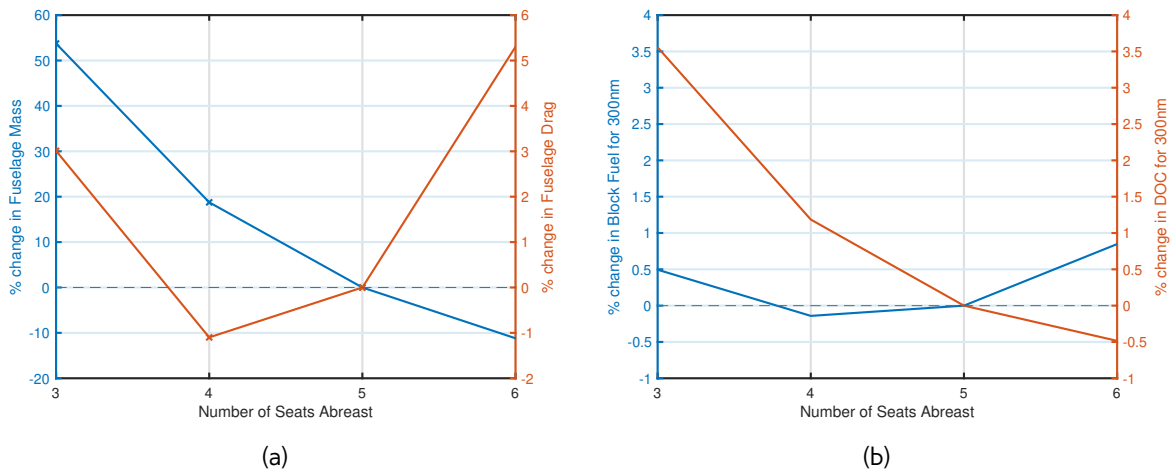


Figure 8: Cabin Sizing Trade-Off Studies

5.1.2 Aerodynamics

The first step of the aerodynamic design was finding a wing area for Falco, which was constrained by the requirements found in the specification [9]. This process led to a wing area of 134 m², which seems large when compared with the competitor aircraft identified previously. This was found to be due to the reasons described in Section 8.1.2. Falco aimed to minimise fuel burn, and so keeping drag low was at the forefront of aerodynamic design thinking. As a result of this, the aspect ratio was an important factor in the design of the wing, and so a trade-off between wing mass and induced drag was carried out and found a value for the aspect ratio of 12.25 to be optimal. With the given wing area, this meant Falco has a large wingspan of 40.5 m. Optimising taper ratio also helps to reduce induced drag by increasing the span efficiency - analysis of this gave a taper ratio of 0.35 which provided the most elliptical lift distribution across the wing. Geometric twist of 2° had to be applied to move the peak C_l location further inboard to ensure control authority could be maintained in the case of stall. Given the low approach speed requirement, combined with the relatively high MTOW of Falco compared with aircraft such as the ATR72, careful design and selection of the high lift devices was necessary to ensure the requirements were met. As a result, leading edge slats and trailing edge single slotted flaps were used, giving a total $C_{L_{max}}$ of 2.54. This value for $C_{L_{max}}$ provides an approach speed of 110kts, which meets the requirement. The flap angle used for take-off was found

by minimising take-off roll, and this resulted in a $C_{L_{to}}$ of 2.00.

5.1.3 Airframe Structure

Falco's airframe structures were designed to withstand combinations of many different loads such as 2.5g manoeuvre, gusts, asymmetric wing loading and internal pressurisation. These loads were quantified from the specification, structures notes and FAA regulations [9–11]. The process involved setting up separate fuselage and wing box load models over a finite number of points and extending these to yield shear force, bending moment and torsion distributions. To size the relevant structural components, such as the stringers, frames and spar, the structures were idealised using a skin-boom approach. Closed cell shear flow analysis was then carried out by making a cut at the point of highest load. An ultimate safety factor of 1.5 was considered such that the structure should not fail until this point. This analysis gave a wing box with 26 ribs, 32 stringers and 8 spar caps per wing. The wing is able to deflect 4.4 m at the tip before material plastic failure, which is greater than the 1.8 m it deflects under ultimate load. The fuselage is comprised of 46 frames, 76 radial stringers and a skin thickness of 1 mm. The material selection processes and justifications are detailed in Section 12, but in summary CFRP is used for the wing box, fuselage sandwich panel floor and bulkheads. Whilst aluminium is used for the fuselage and wing skin. The total bottom-up airframe mass (wings + fuselage) came to 6950 kg.

5.1.4 Architecture and Systems

Falco utilises a more electric architecture to eliminate bleed air from its propulsion systems and improve fuel burn by approximately 1-3% [12]. Designs were made for electrical, hydraulic, data and water systems, with a focus on redundancy and reducing weight. For the electrical system, a normal power generation of 400 kVA was calculated using previous aircraft data, which will be supplied by four 100 kVA VFSGs, two per turboprop. Control surface actuation, de-icing and cabin pressurisation will all be electrically powered, in addition to traditionally electric sub-systems. A hydraulic system was included for safety critical sub-systems such as the landing gear, in addition to redundant actuation of the rudder and elevator. For the data system, an integrated modular avionics architecture was included, in addition to two sets of dissimilar flight computers: ELAC, SEC and FAC. For the water system, a simple system with an emphasis on reduced weight and footprint was designed.

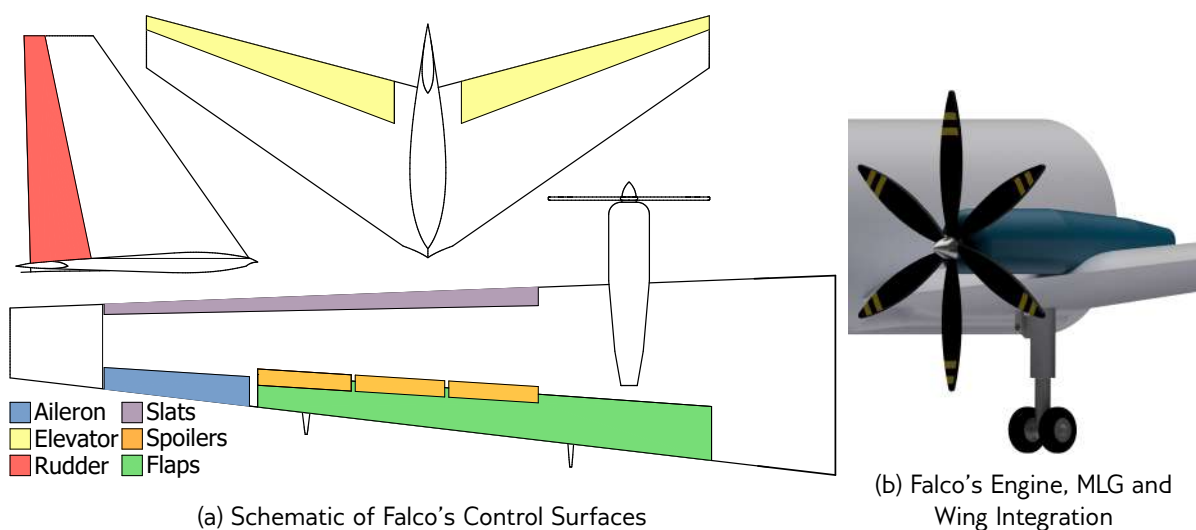


Figure 9

5.1.5 Weight, Balance and Flight Control

Falco has an operational empty weight of 24075 kg, and a maximum take-off weight of 39700 kg. The acceptable centre of gravity range is 6% to 36% mean aerodynamic chord, with a larger acceptable ground loading envelope of 5.8% to 50% MAC to improve operational flexibility. The aircraft features a conventional tailplane layout, chosen due to its low mass and reduced control risk when compared to other empennage designs. Fly-by-wire flight control architecture is implemented as part of two separate electrical systems. Fly-by-wire offers enhanced control and flight envelope protection, which improves safety. Control surface sizing and deflection limits can be seen in Table 1, they were chosen to provide optimal flight control throughout the flight regime.

Table 1: Control Surface Parameters

Control Surface	Total Area [m ²]	Deflection Limits
Aileron	4.4	+25° to -20°
Elevator	7	-25° (up) to +20° (down)
Rudder	4.5	+/- 30°
Spoiler	5.5	40° (flight) 55° (ground)

5.1.6 Landing Gear

The landing gear layout is a tricycle configuration, assembled with 2 radial NLG tyres of size 26x6.6 R14 in, and 4 radial MLG tyres (2 on each strut) of sizes 27.75x8.75 R14.5 in. This configuration allows Falco to satisfy ICAO code C requirements, however, it does not meet the soft requirement of >15° tail strike angle. Scenarios such as failure of NLG were also considered when designing the LG. The shock absorber was also designed and simulated to ensure it would not leak or break. Finally, mass estimation, material trade-offs and preliminary stress analysis were carried out to ensure that the LG would meet the design requirements.

5.1.7 Propulsion and Fuel

An important design decision was the choice between turboprop and turbofan engines. From initial fuel burn calculations, ignoring the mass or aerodynamic differences between the two engines, the turboprop resulted in a 13% reduction in fuel burn over the 300nm study mission. Combined with the improved climb performance, this was significant enough to choose the turboprop as the powerplant for Falco. The engine scaling was governed by balanced field length requirements and the target of low fuel burn. As a rule of thumb, a smaller engine resulted in a lower fuel burn at the cost of SFC efficiency, hence the engine was scaled to meet the requirement of 1500m. An engine scaling of 0.9 was found to produce balanced field length of 1430m.



Figure 10: Falco's Built-In Stairs

5.1.8 Additional Features

Folding wingtips that reduce the wing span to 36 m are an optional feature for airlines with Falco. A trade-off study for the folding wingtips was conducted between DOC increase and airport compatibility, it was found that a slight weight penalty of 190 kg per wing increased DOC by 1%. Research into 30 regional airports in Europe concluded that Falco was incompatible with 11 of these, as they do not have capacity for ICAO code D wing spans. Out of these 11 airports, 5 of these were among the top 25 busiest routes in Europe. Some airports also had up to 20 more gate spaces for 36 m code C spans. This research deemed that Falco should incorporate folding wing tips as an option, to accommodate regional airlines. Built-in stairs are also an option, adding just 91 kg with the potential to cut turnaround time by up to four minutes. The aircraft is equipped with manoeuvre and gust load alleviation to reduce structural fatigue and mass.

5.2 Performance and Footprint

5.2.1 Aircraft Performance

Falco meets all performance requirements, climbing to a cruise altitude of 25,000 ft within 14 mins with an SFC of 0.0413. This results in a total fuel burn of 1170 kg for the 300 nm study mission, 11.2% lower per seat than the current generation ATR72-600. Falco has a BFL of 1430 m, with a maximum climb gradient of 3% to a maximum one-engine inoperative altitude of 16,000 ft. The aircraft has a JAR landing field of 1074 m, allowing for a 50 ft object clearance at a descent pitch of 3°.

5.2.2 Environmental Impact

Falco's cruise altitude is optimised to mitigate contrail formation. With regards to emissions, Falco produces 64g of CO₂ per seat/km, which is 7% lower than the current generation ATR72-600. Falco releases 1.26 kg of NO_x per take-off and landing cycle, which is a 50% reduction in NO_x emissions compared to the ATR72-600. Using statistical equations based on historical data from a wide range of aircraft, a noise estimation was conducted. The results of this can be seen in Table 2, with Falco complying with strict Heathrow noise quotas.

Table 2: Falco Noise vs. Heathrow Quota Counts

Condition	Heathrow quota count (QC) [dB]	Falco [dB]
Sideline at take-off	95.9 (2)	92.0
Approach	92.9 (1)	88.2

5.2.3 Sustainability

Falco's structural materials were chosen with sustainable material lifecycles in mind throughout sourcing, manufacture, usage and end-of-life management. The airframe is primarily made from aluminium, a material that is not only durable, flexible, efficient and low-cost, but also recyclable. Some modern airframes such as the Boeing 787 and Airbus A350 have taken advantage of CFRP's high strength-to-weight ratios and comparatively lower mass than aluminium. However, there are sustainability issues with carbon fibre as it is currently difficult to recycle effectively; a report by the charity Green Alliance stated it could cause serious issues in the future when it comes to waste [13]. On the other hand, aluminium can be recovered and smelted back to its original state, before being re-alloyed and reused.

When it comes to production, Falco will employ sustainable manufacturing practices, such as closed-loop recycling, whereby all offcuts and scrap metal will be reused. Falco will also work with composite recycling firms that divert waste away from landfill and into other products such as electronics and ground vehicles [14].

5.3 Economics

5.3.1 DOC Breakdown

Figure 11 shows Falco's DOC breakdown calculated using the AEA method, giving a DOC of \$5181 per trip for a 300nm mission at full capacity of 120 passengers and a COC of \$3395 per trip [15]. With our design being in its own category for typical trip in terms of PAX and range, when comparing DOC to our competitors it was deemed sensible to compare the seat-mile costs for missions of a range suited for each aircraft as opposed to overall trip cost, as displayed in Figure 12. Falco outperforms its competitors with DOC savings of 15-19% and COC savings of up to 18%. The data shows that Falco is optimised for the given specifications, with lower DOC over the same range, carrying more passengers than the ATR72 and Dash8 Q400. Additionally, the saving compared to turboprops drastically increases over a distance lower than they are suited for, seen by the inclusion of the E190's DOC for a 200 nm trip.

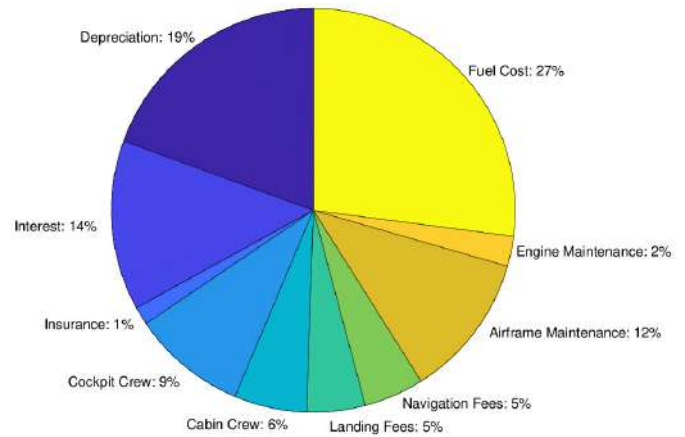


Figure 11: Falco DOC Breakdown

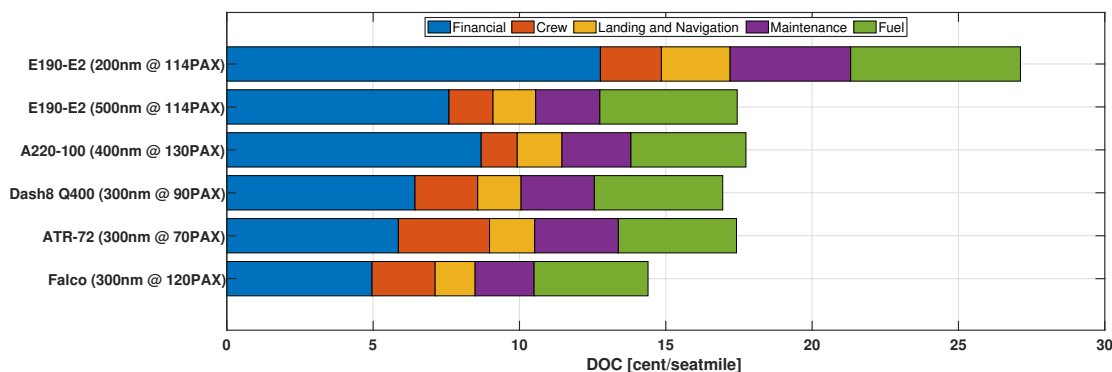


Figure 12: DOC Comparison vs. Competitor Aircraft

To assess the robustness of the economic performance in possible future scenarios, a sensitivity analysis was conducted on DOC for variation of the most volatile factors: MSP and fuel cost. This is shown in Figure 13, where with increasing MSP, Falco is more robust than the ATR72. However, with increasing fuel price we perform worse. The same analysis was carried out again but for COC, where the economic performance is roughly the same up until a fuel price of \$4.5/USgal. It is important to try and mitigate this as fuel may become more expensive with SAFs becoming more common. We should have considered measures to reduce the sensitivity of Falco's economics in relation to fuel price. With hybridisation a growing area of interest, and Falco featuring a more-electric architecture, this could be explored if the technology reaches sufficient readiness in time.

5.3.2 Program Cost and Profitability

Evaluation of the number of aircraft to produce was difficult as no directly comparable aircraft exist. Roskam [16] suggests a market capture of 1/3 is conservative, however, given Falco's superior economic performance, we can be confident in estimating a 40% capture. Applying this to the turboprop forecast of 2450 aircraft as stated in Section 3, we aim to produce 1000 aircraft. Using Roskam's

methods [16] to estimate non-recurring and recurring costs, these came out as \$1.4B and \$23B respectively, and an MSP of \$30M assuming a profit margin of 20%. By summing NRC and RC, and dividing by MSP, the breakeven number was found to be 838 units, well below 1000 demonstrating that large profits can be made from Falco, around \$4.9B.

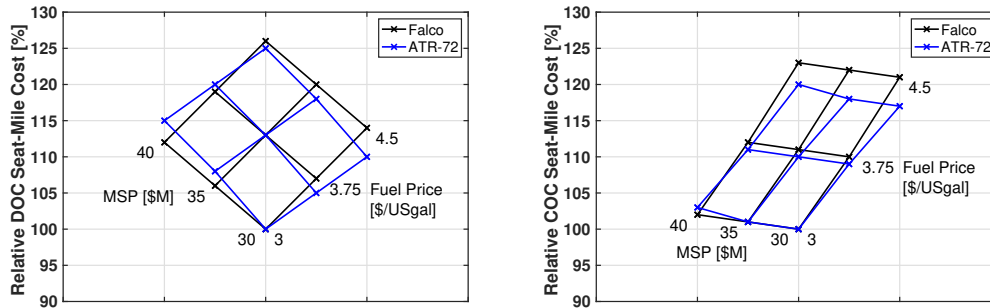


Figure 13: Sensitivity Analysis on (a) DOC, (b) COC

6 AIRCRAFT CHARACTERISTICS VS. SPECIFICATION

Requirement	Units	Specification	Falco	Ref.
Passenger Capacity (1-class)	-	120	120	A
Design Range	nm	800	850	B
Cruise Speed	Mach.	Result	0.47	C
Time to Climb (1500ft to ICA)	mins	<24	14	D
Initial Cruise Altitude	ft	>20,000	25,000	E
Maximum Cruise Altitude	ft	>27,000	30,000	F
Approach Speed	kts CAS	<115	110	G
Balanced Take-Off Field Length	m	<1500	1430	H
Landing Field Length	m	<1250	1070	I
One Engine Inoperative Altitude	ft	Result	16,000	J
VMO / MMO	kts CAS / Mach	Result	357 / 0.54	K
Turn-Around Time	mins	Result	25	L
DOC Target	\$/seat-nm	Result	0.14	M
Expected Entry Into Service	year	2030	<2030	N

Table 3: Falco vs. Specification [9]

- A - Passenger capacity had to be at least 120 PAX. The cabin was sized according to Section 5.1.1, therefore with a 5-abreast cabin layout, 24 rows are required.
- B - The design range was calculated for MTOW, accounting for the maximum possible payload that Falco may carry. Falco exceeds the design range by 6.25% including reserves.
- C - Falco travels at Mach 0.47, optimised between block fuel and block time, whilst allowing the engine to operate at a sufficient surge margin.
- D - Time to climb requirement is met, as the engine sizing scale factor of 90% was driven by the OEI requirements, which means that Falco's turboprops exceed the time to climb requirement.

- E - The initial cruise altitude of 25,000ft meets the requirement, and it was chosen as a balance between minimising the fuel burn of the engine and reducing contrail formation which occurs at slightly higher altitudes.
- F - The max cruise altitude was dominated by the engine performance, which is limited to 30,000 ft.
- G - The approach speed requirement is met, but during the project this was found to be one of the requirements which had a lot of influence over design decisions, especially with regard to the aerodynamics. In hindsight, the team should have clarified how hard this requirement was with the industrial advisors. Approach speed was calculated as $V_a = 1.3 \cdot V_{min}$, more detail on how this requirement affected our design is given in Section 8.1.2.
- H - The balanced field length requirement is met at MTOW accounting for performance margins on MTOW, aerodynamic parameters and SFC. This largely dominated the engine scaling and was calculated using discrete time-stepping methods, verified to 3% accuracy of real data.
- I - The landing field length is met at MLW accounting for performance margins. Again a discrete time-stepping method was used, verified to 2% accuracy with real-world data and accounting for a braking co-efficient of 0.25. The total field length accounts for a 50ft object clearance and a 3-degree approach slope.
- J - The one engine inoperative altitude was a result of the engine scaling and aerodynamic parameters. The increased drag due to windmilling effect of engine and the drag rise from trimming the aircraft was accounted for by methods described by Jenkinson [17]. 16,000ft will give sufficient clearance to fly north over the Italian Alps.
- K - VMO is constrained by structural flutter and MMO is constrained by compressibility effects and flutter. However, as the cruise Mach for Falco is well below drag divergence, compressibility effects are not a concern. The aircraft is structurally capable of flying at Mach 0.54 ($M_c+0.07$).
- L - Turnaround time was calculated to be 25 minutes, using current data for turnaround times as well as passenger loading and unloading rates. Built-in aircraft stairs are included as an option for airlines that wish to have them included. It is estimated that they cut turnaround time by up to four minutes and are ideal for regional airports.
- M - DOC is a result of using the AEA method for the 300nm study mission at 120PAX, using the specified fuel cost of \$3 /USgal [15]. It is at the lower end of typical seat-mile cost for regional aircraft.
- N - see Section 7.1.

7 TECHNOLOGY LEVELS AND RISK ANALYSIS

7.1 Technology Readiness

Gust Alleviation System: The use of gust load alleviation depends on the development of doppler LIDAR sensing systems that can be used to scan the 'wind profile' ahead of the aircraft in flight [18]. Whilst manoeuvre load alleviation can already be seen on aircraft such as the 787, the technology readiness of the LIDAR detection system poses a risk to the implementation of gust load alleviation. In the past 20 years much development has been carried out as part of the European CleanSky Joint Technology Initiative – however it is speculative whether the system will be available before the launch of Falco in 2030. Funded by the Cleansky Initiative, the GLAMOUR project has particular relevance, with its aim to progress gust loading control algorithms on a 100 seat passenger aircraft [19].

As the aircraft is equipped with manoeuvre load alleviation ability, implementation of gust load alleviation could be offered as a retrofit once available, requiring only the addition of the LIDAR sensing system to the nose of the aircraft due to the modular nature of the avionics.

E-taxi: The E-Taxi technology was deemed unfeasible for Falco's design as it demanded an increase in weight (at least 150-200 kg) [20], and a huge purchase cost (around \$1.5 million [21]), which doesn't

align with Falco's design drivers. Currently, the additional weight and costs of this technology outweigh its turnaround time benefits. Future improvements could make it viable, however, at the moment this is not the case.

Overall Development Costs: The R&D costs (NRC) were estimated using Roskam's method, which accounts for the costs of airframe engineering and design for the suggested 5 airframes, of which 2 are developed to production standard for flight tests with these costs also accounted for. Finally, there are the costs of financing the R&D phase, with a 10% profit margin added. One of the most influential factors is the difficulty judgement factor, for which Roskam suggests between 1 and 2. While our design is fairly conventional, 1.8 was chosen to be conservative given that we are incorporating folding wingtips and a more electric systems architecture both of which are still recent developments, seen in the 777X and 787 respectively. With our wing being smaller than the 777X this means the structural complexity will be lower, and with more electric systems already having a heavy focus on their further development, this gives us confidence that Falco can be developed in the specified timeline with entry into service in 2030.

7.2 Risks and Mitigation

All engineering endeavours carry a certain amount of risk and Falco is no different, with both design-specific and project-completion risks being present.

CFRP Considerations: Falco utilises a lower amount of advanced modern materials compared with some of its competitor airliners, for example, composites make up more than 40% of the A220s primary structure [22]. This poses the risk of Falco being outclassed in the next few decades by next-generation lighter more fuel-efficient aircraft. However, there are issues with small impact damage and its detection; a U.S Government report highlighted how impact damage to composite structures may not be visible [23], making it more difficult to detect than damage to metallic structures. The choice of aluminium for the airframe is also justified by CFRP's current poor sustainability during manufacture and at end-of-life.

The wing box contains CFRP stringers and spars connected to its aluminium skin, posing thermal expansion and galvanic corrosion issues. These are mitigated in Falco's design, through the use of thin fibreglass layers between these boundaries which electrically isolate the aluminium and CFRP.

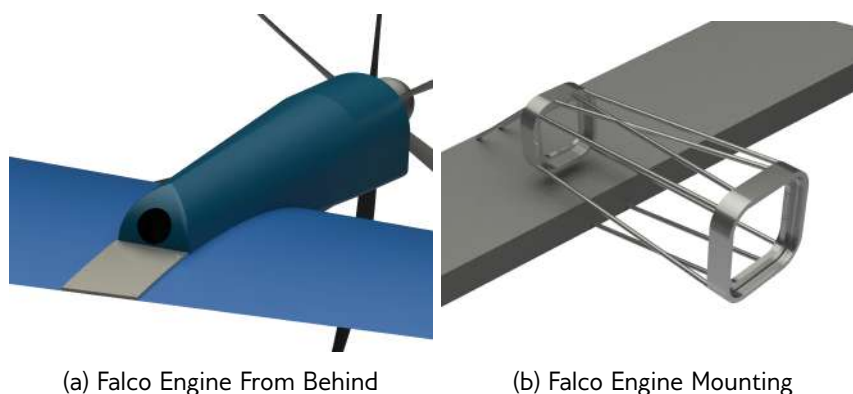


Figure 14

Engine Placement: The integration of the powerplant within the wing ignited a few design complications. It is partly integrated and not mounted using a traditional pylon system common with turbofans, where the nacelle is restricted to a cylindrical shape. To reduce root twisting on the wing box, the engine is mounted on both the front and rear spar, sharing 75% and 25% of the load respectively. The mounting structure can be seen in Figure 14b. Beyond this, there is the issue of hot exhaust

gases over the aft region of the main wing. To eliminate any damage due to thermal loads, the rear of the nacelle and the aft strip of the wing is fitted with a titanium heat shield. This can be seen in Figure 14a, inspired by the design of the Avro 748 [24]. Titanium was chosen due to its excellent thermal properties; the additional material cost is outweighed by the potential maintenance cost of alternative materials. The mounting structure allows for sufficient blade clearance in the event of NLG collapse, whilst additionally benefitting from increased lift over the main wing.

The engine is mounted close to the fuselage to reduce the resultant moment during engine failure, however, this inevitably leads to a loud cabin. To mitigate this the aircraft will use passive techniques; the void between the cabin walls and fuselage skin will be filled with materials which exhibit high sound absorption coefficients, such as mineral wool commonly seen in aviation [25].

Aircraft Delivery: There are some inevitable risks with large-scale aircraft manufacture, ranging from supply chain issues to fluctuations in material and energy costs. Although some of these are unavoidable and unfortunately out of our control, we have attempted to design Falco with these risks in mind. Our landing gear and engines use titanium, which is currently facing large supply chain issues. Whilst not known for certain, multiple reports suggest that up to 65% of the titanium used by Airbus comes from Russia, whose exports are currently under sanction due to the conflict in Ukraine [26].

Project Delivery: In terms of project risks, there was an ongoing chance that one or more team members fell ill, hindering progress. As a team, we deemed this fairly likely and quite severe due to the project only being around 10 weeks, with each person overseeing a specific discipline. To mitigate this, the project manager produced detailed meeting minutes and action points, such that ill team members could stay in the loop and begin working from home. We ended up facing this issue on two separate occasions, impacting our ability to make progress. However, the mitigation we had in place meant another team member could lead the meeting and follow the notes and action points format. In addition, all files were stored in a Microsoft SharePoint site so absent members' data could be accessed by everyone else. This also mitigated the risk of project data loss as all files were stored in the cloud.

8 CRITICAL ANALYSIS OF DESIGN AND WAY OF WORKING

8.1 Critical Analysis

8.1.1 Conventionality

Although our concept selection process incorporated unconventional designs, it resulted in a traditional aircraft configuration being chosen, which was backed up by a rigorous sensitivity analysis. However, this means that Falco looked similar to some older aircraft, such as the Avro 748 and BAE ATP. Considering other alternative configurations, or reassessing the weightings in the ranking matrix, could have resulted in a more unconventional aircraft being pursued. However, it should also be noted that civil aviation focuses on incremental improvements. Whilst we are aware that such a conventional design leaves us vulnerable in the market to similar aircraft; we believe that airlines are more likely to purchase an aircraft they are familiar with and that passengers would be comfortable with.

In hindsight, Falco lacks the implementation of novel technologies. Whilst the optional folding wing tips and inbuilt stairs improve airport compatibility and airline acceptability, there are no innovative systems that directly reduce fuel burn and thus the environmental impact. If conducting the project again, we would definitely research and utilise new technologies that could help Falco to reduce its emissions.

8.1.2 Overall Wing Design

One aspect of Falco that has been spoken about among the team as well as during the FDR is the oversized wing. The wing has an area of 134 m^2 , which seems very high when compared to an Airbus A320, which has a 34% higher MTOW and a wing area of only 122 m^2 . There were several reasons identified for this high area. Firstly, as the detailed design phase went on, more accurate mass estimations meant the MTOW decreased (Figure 15), but the wing area had already been frozen and so the wing was sized for a slightly heavier aircraft. Furthermore, the 115 kts approach speed requirement was quite low for a large aircraft, a Boeing 737 has an approach speed of 140-150 kts, for example. This meant that the design stall speed was very low, and so a high wing area was required to achieve this low-speed performance. Another reason for the large wing area is that when converting approach speed to stall speed, the old FAA factor of 1.3 was used rather than the 1.23 used for fly-by-wire aircraft. This meant that the stall speed that Falco was designed for is 10% lower than it needs to be according to FAA regulations. All of these factors resulted in a wing that looked very large compared to the overall aircraft size, and this ultimately had an effect on the aerodynamics of Falco.

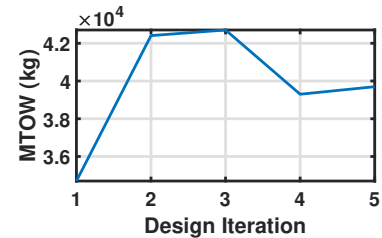


Figure 15: MTOW Iteration Progression

8.1.3 Integration of the Landing Gear and Wing Box

Part of the bottom skin of Falco's inboard wing box is cut out to make space for the main landing gear, weakening the structure. This was an issue that did not occur to the team until a couple of weeks before the FDR, somewhere along the line we hadn't considered it. This was an effect of having an unswept wing with no place aft of the wing box for the MLG to be stored. It was too late to relocate the MLG and so we decided to go about implementing mitigation. This came in the form of reinforcement with additional stiffened stringers and a thicker skin at the section, to withstand excessive torsion. The fuel tanks are integrated outboard of these cutouts as fuel volume is not an issue, also providing some wing root bending moment relief under load. This issue is something that should have been spotted during a design meeting. In retrospect, we would hold more regular and detailed reviews of the general arrangement drawing as we sometimes did not take full advantage of its potential as a design tool.

8.1.4 Economic Assumptions and Future Risks

It should be noted that methods used contain many assumptions, like constant labour rates in the AEA method that in reality will change with inflation. Also, many of the costs used in Roskam's methods are from 1970 and 1989 which had to be adjusted using inflation to today's prices which may not be fully accurate. The method contains multiple judgement factors as well, such as program complexity, level of CAD experience, level of simulation technology required, and material usage. These are difficult to estimate and in the case of program complexity have a large effect on the outcome, thus reducing the accuracy of the cost estimations. Furthermore, future market demand is hard to predict and can swiftly change with geopolitical or population factors, and the most desirable routes can also change, along with fuel price and market inflation.

8.2 Way of Working

Roles: Before the PDR, team members worked on different aspects of the design to get a good idea of what each discipline entailed and how they tied together with the overall process. This led to each of us naturally working on the area we had the most relevant skills for and were most passionate about. This meant that after the PEDR, when it came to allocating disciplines, everybody was able to work on their first choice, except for one person who had their second choice.

Organisation and Process: We had compulsory, bi-weekly meetings at recurring times that fit into everyone's timetables, along with irregular smaller group and one on one meetings where necessary. The big team meetings were led by the Project Manager, who ensured notes were taken and action points with soft/hard deadlines were given out. The PEDR proposal acted as our first design, a starting point upon which we iterated. We worked to a development model that involved weekly freezing of sizing and data, from which we reevaluated the mass breakdown, performance metrics, environmental impact and economics of our iteration. The next iteration involved resizing and recalculating based on this frozen sizing, data and analysis. In total, we performed five iterations (Figure 15), each time reviewing the data against the specification and discussing improvements.

Microsoft Excel was used to store all data values from each discipline in one place, this allowed any team member to see the latest data without having to contact the relevant person. The Excel workbook was guarded by a change control system, in which team members would log each time they made an update to their values. Separate sheets were used for each iteration and its relevant mass breakdown. All of the team's files, including this iteration workbook and the GA, were stored in a Microsoft SharePoint Site connected to a Teams group. The Teams had channels and folders for each discipline, making locating files easier, with the 'General' folder used by the Project Manager to store meeting notes and work for the assessed work.

Falco Change Control Document							
Request No:	Request Date:	Requested By:	Discipline:	Documents for Request:	Description:	Additional Comments:	For PM: Approved?
#000 (example)	31-Oct	Bob	Aerodynamics	Aircraft Data Sheet v2	Updated wing parameters (AR, span, taper, sweep) for v2 iteration	based on new MTOW = 42407 kg	in process
					FROZEN - Aircraft Data Sheet v1		
#001	01-Nov	Charlie	Structures	Aircraft Data Sheet v2	input v2 fuselage parameters	unchanged as fuselage structures work still low fidelity	yes

Figure 16: Change Control Tracker with Example

Lessons Learned: Reflecting on our way of working, it is clear that our iteration method did not work as well as intended. At the start and end of the process, it was very smooth and there was little confusion in finding up-to-date, correct data. However, a couple of weeks after the PEDR most team members needed more time to refine their methods and perform some more in-depth analysis, it felt like constant resizing was getting in the way. This led to us deciding to freeze the wing size earlier on than we should have, as explained in Section 8.1.2. In hindsight, resizing based on a frozen set of data and mass meant the iterations never really caught up with each other. The more detailed analysis, required for FDR, took time to set up and it was too much work to resize every major component every week. Next time we would attempt to do major iterations of a few components at a time, rather than everything at once. For example, resizing the wing aerodynamics and engine sizing, before recalculating loads and then resizing the airframe accordingly. This would lead to smaller, more manageable changes in MTOW, having less of a domino effect on the next iteration.

9 SUMMARY

This report presents the design of a 120 passenger, 800 nautical mile range airliner. The aircraft is configured with a low wing, conventional tail and twin turboprops. This design was chosen from a detailed concept analysis which considered a variety of aircraft arrangements. Falco was designed around a philosophy of minimising mass and maximising aerodynamic efficiency. The design meets all the requirements set out in the specification and achieves our design drivers of minimising environmental impact and maximising airline and passenger acceptability. In this report the market analysis, concept selection and description of final design were covered, along with an assessment of the aircraft's environmental impact and technology readiness. Risk level and critical analysis were also considered. The report concludes with more in-depth technical reports for a range of specialisms covered in the AVDASI4 project.

DESIGN DETAILS

10 SYSTEMS

Initially, a top-down approach was taken to the systems of Falco. Research was carried out to determine which types of system would be incorporated and designed, with electrical, hydraulic, data and water systems being selected. The omission of a pneumatic system was deliberate, with the objective being to eliminate bleed air from the propulsion systems. Depending on flight conditions and aircraft, the use of bleed air can have a fuel burn penalty of 1-3 % [12]. This led to the choice to make Falco a more electric aircraft based on the aircraft's primary design drivers. Continuing the top-down approach to Falco's systems, a general systems arrangement (Figure 17) was made with a focus on redundancy and safety. Once the major parameters of each system had been decided, a bottom-up approach to each system could be carried out, where individual sections of each system could be sized and selected.

10.1 Electrical

To initially calculate the power budget required for the electrical system, the power budget of other more electric aircraft was scaled based on MTOW. This approach gave a total power budget of 400 kVA. Based on this value, the primary method of power generation for Falco is four 100 kVA VFSGs, with two VFSGs being driven by each turboprop. A similarly sized generator has been used in the Airbus A350 XWB, which utilises 4 Hamilton Sundstrand 100 kVA VFGs [27]. To provide power for turboprop start-up, an APU was included within the electrical system. The Honeywell 131-9A is used on the Airbus A320 family and Airbus A220, and as such was deemed suitable for Falco. Batteries are also an important feature of a MEA, and like other similarly sized aircraft Falco has a main battery and APU battery, which both use lithium-ion cells. Issues resulting from batteries have been shown previously, such as the Boeing 787s use of lithium-ion cells resulting in batteries overheating. However more stable chemistries have become available and adequate insulation will be given to the cells to mitigate the risk of over-heating. A RAT assumed to be approximately 800 mm in diameter was included for emergency power generation.

With regards to electrical distribution, two TRUs were included in the design of the electrical system, in combination with 28 VDC buses and 115 VAC buses. Additional buses may be necessary, however this will be determined by sub-systems choices made in future iterations. The primary TRU is responsible for converting 115 VAC supplied by the turboprop driven generators into 28 VDC for batteries and DC components. The secondary TRU is responsible for converting 28 VDC from the APU Battery into 115 VAC for APU start-up. The secondary TRU can also act as a back-up in the event the main TRU fails.

Future iterations of the electrical system of Falco would allow for more detailed analysis of its electrical subsystem's. The choice of electronic actuation would require further work into the implementation of EHAs, with a focus on redundancy and weight savings. The lack of bleed air means that sub-systems that are traditionally pneumatic must be powered electrically. One such system is the aircraft's de-icing system, which on regional aircraft typically takes the form of de-icing boots along the leading edge of the wing or the circulation of hot bleed air to heat surfaces. For Falco, electric heating elements will be used for the leading edge and propellers for the purpose of de-icing and anti-icing. This approach would allow for localised de-icing, meaning only areas where ice is building up are heated, leading to reduced power loads. Similarly, cabin pressurisation is typically achieved using bleed air. Falco's ECS will utilise electrically driven compressors, with air intakes integrated into the aircraft's belly fairing.

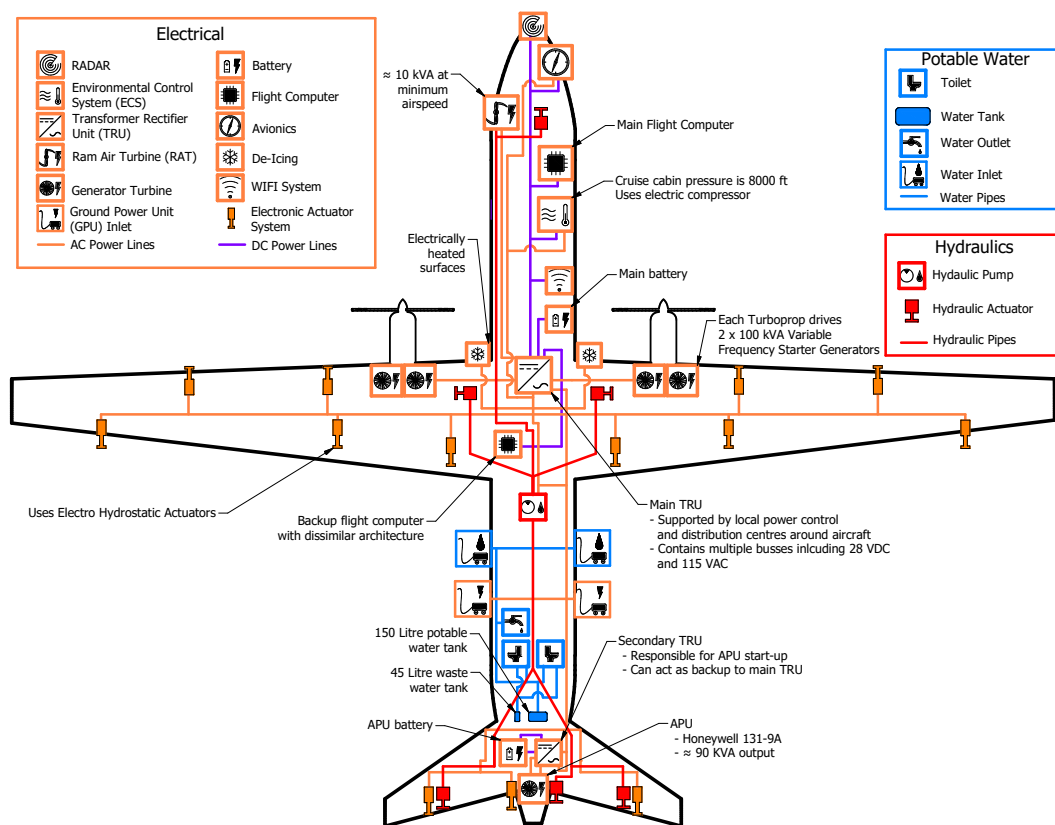


Figure 17: General Systems Layout

10.2 Hydraulic

The overall mass of a hydraulic system tends to be greater than an electrical equivalent, despite the fact that hydraulic actuators tend to be lighter than electro-hydrostatic actuators for equivalent performance. The higher mass of the overall system is a result of the greater weight of hydraulic pipes compared to electrical cables, in addition to the hydraulic pumps and motors needed for the system to function. However, hydraulic systems within aircraft are a proven and mature technology, and as such have been included in Falco in safety critical systems and for redundancy.

The MLG and NLG are hydraulically lowered and raised as they are both safety critical and the mass of electric actuators needed to replace hydraulics would be impractical. Hydraulics actuators have also been included within the design to provide redundant control of Falco's elevator and rudder in the event of electrical system failure. The hydraulic system is powered by an electric motor, and in emergencies the RAT can also provide hydraulic power.

10.3 Data

The primary design drivers for Falco's data system were safety and future readiness. As such two flight computers were included within the data system, with the secondary computer featuring a dissimilar architecture and acting as a back-up. Optics were considered for a fly-by-optics architecture, however their immaturity, problematic maintenance qualities and the need to redesign traditional data architectures were significant enough drawbacks to disregard the potential weight savings.

With regards to avionics, the Falco utilises an Integrated Modular Avionics (IMA) architecture. Modularity allows for both simpler upgrade paths and reduced maintenance costs through easier isolation of issues.

10.3.1 Flight Control Architecture

To reduce the risk of overall failure, flight control is split between two separate and dissimilar electrical systems, A and B. A hydraulic actuation system provides redundancy in the event of system failure. Each electrical system contains one set of flight computers (ELAC, SEC, FAC), which can both independently control the aircraft.

Table 4: Non-Exhaustive List of Falco's Avionics

Avionic Subsystem	Supplier	Name	Description
Traffic Collision Avoidance System	Thales	T3CAS	Also includes Terrain Avoidance Warning System
Air Data Inertial Reference Unit	Thales	TopAxyz	
Global Navigation Satellite System	Thales	Topstar C	Compatible with Galileo
Flight Management System	Thales	PureFlyt	
Health and Use Monitoring System	SAFRAN	ACMS	
Weather Radar	Collins Rockwell		Overflight weather protection included

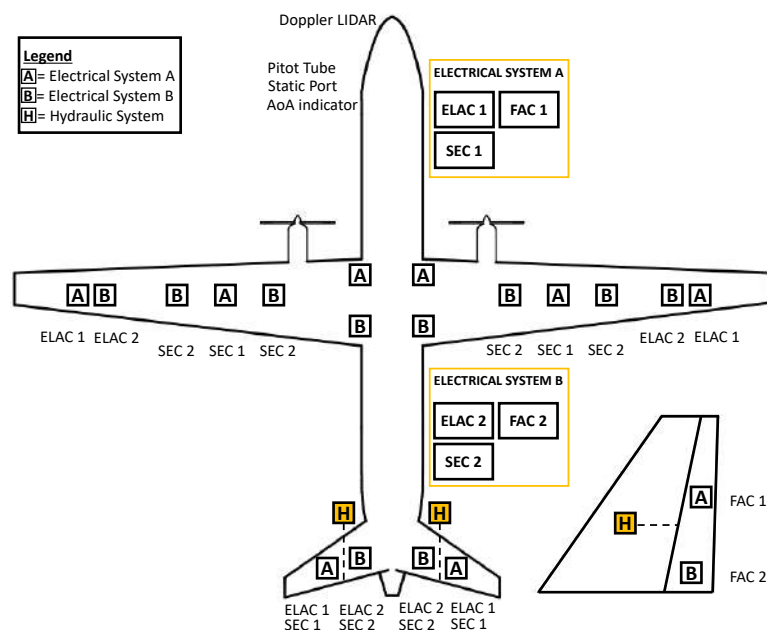


Figure 18: Flight Control Architecture

10.4 Water

Falco's water system was designed to minimise weight and footprint. To achieve this, the water system was concentrated within the rear of the aircraft to reduce the mass of pipes needed. A 150 L potable water tank was included in addition to a 45 L wastewater tank, to give a total water system mass of 210 kg. The inclusion of water treating sub-system was considered for Falco. However, upon further evaluation, it was found that the short range and flight time of Falco's design mission resulted in no improvement to fuel burn. This was due to the increased weight of equipment necessary for water treatment being greater than the reduction in water needed. The increase in fuel burn was also deemed more important than the environmental benefits of reduced water requirements.

11 WEIGHT, BALANCE, STABILITY AND FLIGHT CONTROL

11.1 Weight

Table 5: Full Aircraft Mass Breakdown

Component	Mass [kg]
Structure:	
Wings	3540
Fuselage	3400
HTP	392
VTP	407
Engines (inc. pylons)	4580
Landing Gear (total)	2434
Total	14753
Systems & Avionics:	
Bleed Air System	150
Fuel System	230
APU	164
Hydraulics	900
Air Conditioning	700
De Icing	42
Fire Protection	90
Flight Controls	350
Instruments	80
Autoflight System	130
Navigation	500
Communication	277
Electric Generation + Distribution	1400
Total	5013

Component	Mass [kg]
Operator's Items:	
Oil	95
Water	228
Toilet Rinse	72
Toilet Precharge	36
Aircraft Documentation	46
Passenger Seating	1800
Trolleys	125
Fixed Equipment	120
Catering	360
Other:	
Doors	66
Aircraft Dependent	50
Passenger Dependent	36
Flight Crew	445
Total	3479
Standard Passenger Payload	11400
Maximum Fuel Capacity	4250
Start up & taxi out fuel	-25
Basic MWE	19765
Evaluation MWE	20360
Basic OWE	23840
Evaluation OWE	24075
MZFW	35475
MLW	37960
MRW	39725
MTOW	39700

Where possible, component masses were estimated using structural volume and material density. This was conducted for all structural components, with the exception of the vertical and horizontal stabilisers. Masses of 'off the shelf' components, such as flight deck instruments and the APU, were found from real world values [28]. For components with no equivalent real-world data available, mass was estimated using empirical methods from a range of textbooks [17, 29, 30], with the results compared to assess the accuracy of the prediction. This was performed for flight controls, VTP and HTP.

Early in the design process a 'bottom-up' mass approximation was performed, which surprised the team with a MTOW 8 tonnes heavier than the previous mass estimate. This was a vital step in the project as it allowed the aircraft to be designed around a much more accurate weight value, preventing late correction of undersized components. MTOW was constantly evaluated and updated throughout the project (see Figure 15) with various 'design freezes', which allowed the team to work to the same mass value for their calculations.

11.2 Balance and Stability

With airline acceptability a key design driver, aircraft loading flexibility was considered early in the process, with the loading diagram seen in Figure 19. Handled in a similar manner to weight data, the aircraft CG position was continually tracked throughout the project to ensure its position within an acceptable window. For ground operations the CG limits were calculated with respect to nosewheel load factors. The rear ground CG position was determined to ensure a minimum of 2.5% MTOW on the nosewheel, ensuring directional stability when taxiing and taking off. The forward CG was dependent on the maximum permissible nosewheel ground load, calculated by the landing gear specialist.

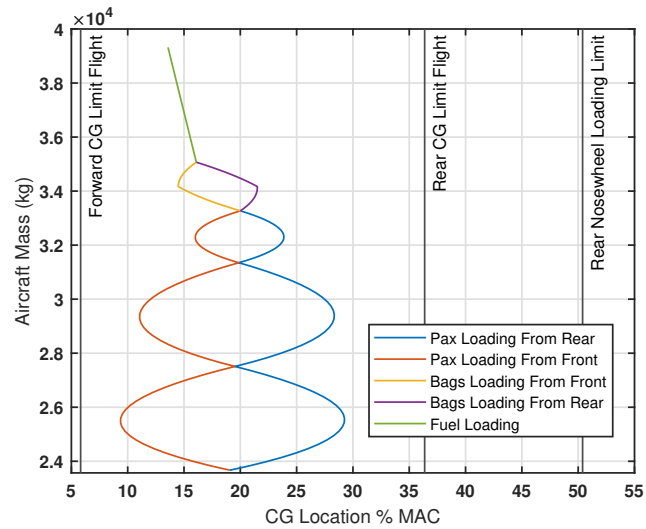


Figure 19: Aircraft Loading Diagram

In flight stability was governed by the aircrafts neutral and manoeuvre points, and the maximum nose up authority of the elevator to provide the required pitch control throughout the flight envelope. These were calculated according to [29, 30].

After the wing design was frozen, the CG was actively managed to ensure it remained in an acceptable region. To adhere to the disk burst condition the engines had to be moved forward by 1 metre, which shifted the CG forwards. To compensate, the position of the main electric distribution system was shifted rearwards to reduce the nose heavy CG position.

11.3 Flight Control

11.3.1 HTP and VTP Design

Design optimisation of the horizontal and vertical tailplane was performed using VLM methods, with reference to current aircraft data for the initial starting point and verification. Aspect and taper ratio were chosen to offer the highest spanwise efficiency. Although there is no aerodynamic benefit at low Mach numbers, sweep was applied to both the HTP and VTP to shift the aerodynamic centre aft and increase the tail moment arm. This resulted in a lower structural mass increase than enlarging the fuselage or tail area to achieve an equivalent tail moment size.

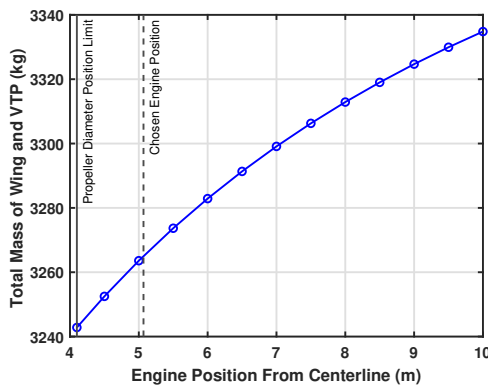
A trade-off study, seen in Figure 20a, was conducted to assess the impact of engine positioning on overall wing and VTP mass. Placing the engines further outboard reduces the wing root bending moment but requires a larger VTP to ensure directional control during a one engine inoperative scenario. The lowest overall mass was found for engines situated closer to the fuselage. As a result the engines are located so that the blade tips are around 1 m away from the fuselage, in order to avoid fuselage impact from vibrational or impact movement and prevent the propeller entering boundary layer flow.

11.3.2 Control Surfaces

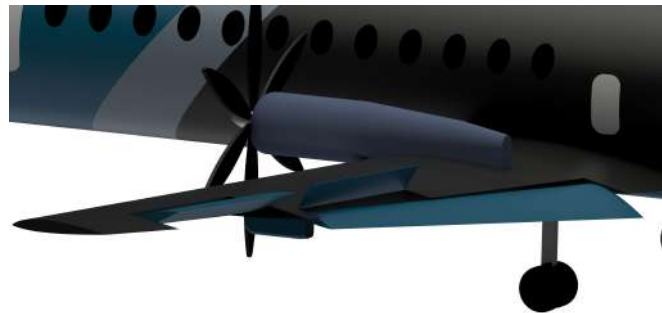
The flight controls were designed according to the CS-25 requirements [31]. To provide full control authority at the low approach speed the ailerons are relatively large, which provides sufficient roll rate

at the cost of slightly increased mass. The maximum roll rate of 34 °/sec (at approach speed with full flaps) was considered too high and was electronically limited by fly-by-wire to reduce structural load and fatigue. Full sidestick deflection commands a roll rate of 15°/sec.

With use of load alleviation technology, the large size of the ailerons does benefit the overall reduction in wing root bending moment. During periods of high aircraft load, both ailerons deflect upwards, shifting the lift distribution inboard and reducing the load on the outer wing. It was found that the mass penalty from larger than required ailerons was extremely low, and the potential reduction in mass and wing fatigue from the increased load alleviation capability of larger ailerons was much higher.



(a) Engine Position Trade-Off



(b) Wing Control Surfaces

Figure 20

The elevator provides a pitch rate of 7° per second during take-off rotation. For our performance envelope we found that using the fly-by-wire system to automatically adjust the elevator deflection to trim the aircraft is more aerodynamically effective than the use of a trim tab or all moving tailplane. The rudder offers a 'best in class' crosswind performance of 37 kts, a higher limit than other comparable aircraft such as the ATR72. The rudder minimum control speed also more than meets the CS-25 requirements and is substantially lower than our stall speed [31]. This improves directional control in the event of an engine out scenario. Wing spoilers were included to allow an increased descent rate in flight, and to reduce lift, increase drag and improve braking performance after touchdown. The spoilers are also used alongside the ailerons to roll the aircraft. Spoiler placement was limited due to propwash – spoilers situated in the airflow behind the propeller would experience unnecessary strains on the actuators.

11.4 Reflection

The fly-by-wire control philosophy was found to be extremely beneficial. Fly-by-wire not only improves aircraft safety by implementing flight envelope protections, but also allows the use of control enhancing features, like load alleviation mentioned previously, and directional correction, automatically deflecting the rudder to counteract the yawing force generated by the P-factor of the propellers.

The trim method selected had implications on our redundant control systems – without the ability to use a mechanical linkage between the flightdeck trim control and a trim tab or all moving tailplane, use of a hydraulic back up actuator was necessary for elevator control. For future work a fuel burn trade-off will be conducted to compare the improved aerodynamic efficiency with the increased mass of a backup hydraulic system.

12 STRUCTURES

To begin the airframe structural design, it was necessary to draw up the flight envelope shown in Figure 21, which is a combination of the specification, structural requirements and airworthiness regulations. The structure was required to withstand loading at load factors of $n=-1$, $n=1$ and $n=2.5$ [10], in combination with gust values taken from part 23 of the FAA regulations [11]. V_d and V_c were calculated from the specification [9], whilst the positive and negative stall characteristics were calculated from $C_{L_{max}}$ using the relationship $nW_{MTOW} = \frac{1}{2}\rho V_s^2 S C_{L_{max}}$. This resulted in the maximum expected load at $n=2.72$, the structure shall not fail until the ultimate load is reached, for which a safety factor of 1.5 was used.

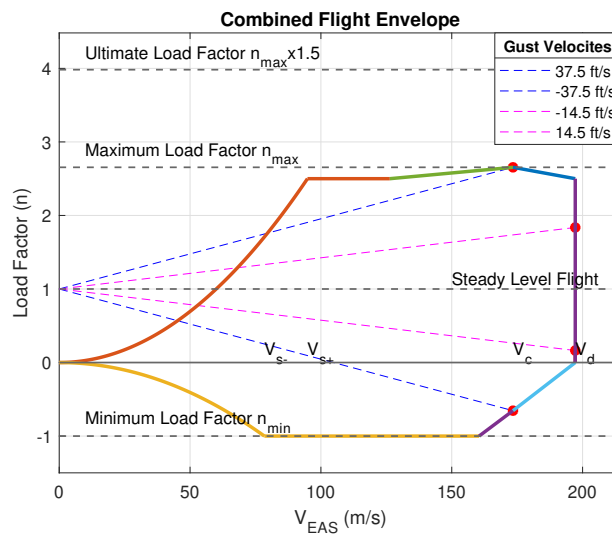


Figure 21: Combined Flight Envelope

For structural analysis, the wing was modelled as a semi-monocoque torque box with two spars. Stringers were used in the upper and lower skins of the wing box. The leading edge and trailing edge spars were positioned at 12% and 65% of the wing chord, to accommodate the control surfaces. The height of the rectangular wing box was assumed to be 90% of the aerofoil height at the leading edge.

From skin-boom idealisation and buckling analysis, the number of stringers and their cross-sectional areas were calculated: 16 on the upper and 16 on the lower skin. The total spar cap area per corner was calculated from this. It was decided that there would be two spar caps at each corner. This design choice was made in order to reduce bulky spar caps and to distribute the bending stresses between two caps for reduced fatigue stress. The stringers and spar caps are made from CFRP, whilst the ribs and skin are made from aluminium 2024; the material selection process is explained later in this report. Z-stringers, with 3.25mm thickness and 20 mm length, were chosen as they allow for easy access to fasteners during manufacture and maintenance. Further analysis using closed cell shear flow methods was conducted to obtain wing box skin thickness and spar web thicknesses. A constant wing box skin thickness of 3.2 mm throughout the span and web thickness of 4.25 mm was calculated. For further work, the skin thickness could be varied along the spanwise direction, to aid in the reduction of overall wing mass. Utilisation of buckling analysis showed that 26 ribs were required per wing. The wing box parameters calculated were comparable to the A320 and B737 due to the similar geometry of the wings [32,33].

Folding wing tips are an optional feature, subject to airline choice. The mechanism added a weight penalty of 190 kg per wing [34], modelled as a point load at 18 m in the wing box analysis. The wing tip was introduced between the last two ribs and was modelled as a 'dummy' structure. The tips fold

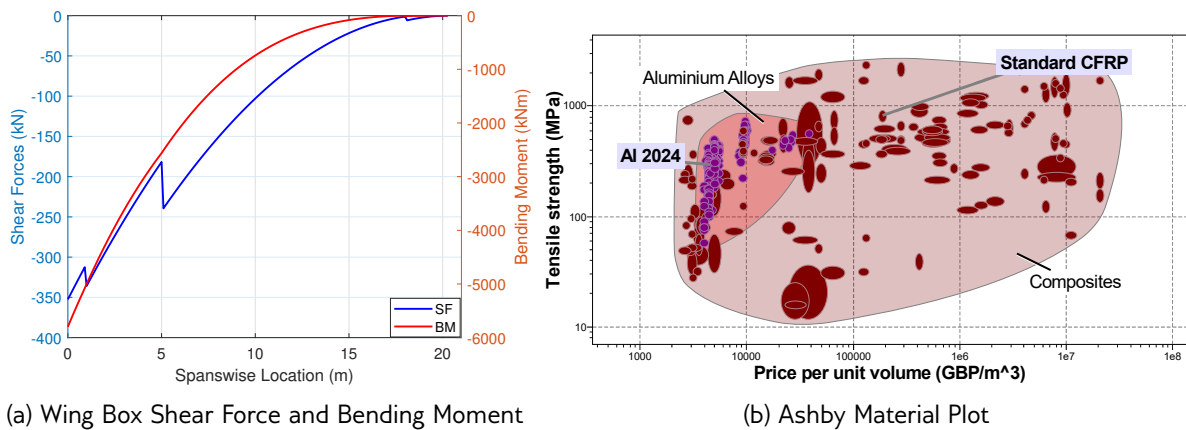


Figure 22

whilst Falco taxis to parking and extend to lock while taxiing to the runway. The folding mechanism was designed to withstand all regulations set by the FAA and EASA to be operational. The mechanism is electronically powered; it was not investigated in-depth as it was deemed beyond the scope of this project.

From the Ashby plot, shown in Figure 22b, it can be seen that Standard CFRP has higher strength but costs much more than aluminium. The loads on the wing were calculated and an initial analysis using aluminium for stringers and spars was conducted. The initial analysis indicated that the stringers and spar caps would be around 1400 kg heavier with aluminium and provide a lower strength-to-weight ratio. With Standard CFRP, the wing could withstand ultimate load at a lower weight. Due to the wing being oversized, heavier stiffeners would be detrimental to the aircraft structure during the aircraft lifecycle. Hence, Standard CFRP was selected as the material for the stringers and spars.

From the wing analysis, it was estimated that the volume ratio of stringers and spar caps to ribs and skin was 2.5. For the skin and rib material, aluminium alloy 2024 was chosen, because the cost of making the skin from aluminium is much lower than from CFRP per unit volume. Aluminium 2024 has high shear and yield strength, as well as good fatigue life, which was deemed important as airlines prefer to spend less on maintenance and reworking of aircraft parts. The use of aluminium and CFRP together leads to integration issues as discussed in Section 7.2.

Before starting the fuselage structural design, its diameter and length were estimated from the cabin dimensions (known from initial cabin sizing) and empirical data given by Nita and Scholz [8]. A longitudinal model of the fuselage was created using a distributed mass model and the wing and tailplane lift forces. The mass model started with low fidelity including point loads; as the design became more detailed the model was updated to become an accurate combination of many distributed loads, based on the locations of major components, systems and payloads. During the analysis, combinations of manoeuvre loads with gusts, internal pressure at maximum altitude and anti-symmetric wing lift loads were considered. The models were initially setup with arbitrary material properties and once refined, material selection was considered.



Figure 23: Fuselage Structure Cut-Out (not to scale)

Shown in Figure 24, shear force and bending moment diagrams were constructed. The largest shear

force and bending moment were found as expected at the centroid; these are given in the summary Table 6. A 'cut' was made at this point and the cross-section idealised using a skin boom approach, with the assumption that the skin thickness and boom areas were equal around the section. A stringer number of 76 was used, scaled from the A320 [35]. The direct stresses in each stringer were found using the relationship $\sigma_{z_i} = \frac{M_x}{I_{xx}} \cdot y_i$. Equating the maximum stress at ultimate against the material proof stress, gave boom areas of 242 mm². Top hat stringers were decided on due to their great torsional rigidity, bending stiffness, and buckling resistance. Shear flow analysis with a closed cell constant was then carried out, yielding a skin thickness of 1.0 mm. To analyse the frames a buckling method by Shanley was used [36], in which the structure is designed to prevent the skin-stringer combination from buckling, making the frames stiff enough to prevent global buckling failure. The aim was to select a stiffness, spacing and cross-sectional area to achieve this with minimum mass, which after iterating were found to be 46 frames at a spacing of 670 mm.

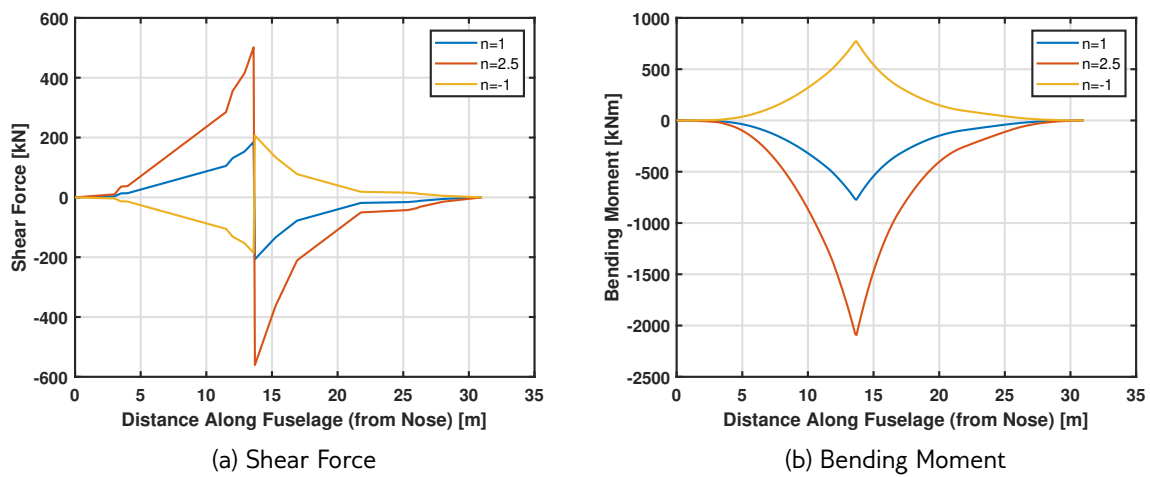


Figure 24: Result of Calculations Along the Fuselage Length

It was decided that the airframe be made of aluminium and not CFRP, due to the reasons given in Sections 5.2.3 and 7.2. Aluminium is a very widely available and recyclable material, meaning its price is stable and will not provide a bottle neck in the supply chain. The aluminium alloy 2024 was selected due to its high yield strength, it offers better workability and fatigue resistance than 7075 which was the other main candidate [37].

Table 6: Summary of Fuselage Structural Values

Max. SF	Max. BM	Δ Pressure	Boom Area	Skin Thickness	Stringers	Frames	Mass
<i>kN</i>	<i>kNm</i>	<i>N/mm²</i>	<i>mm²</i>	<i>mm</i>	-	-	<i>kg</i>
-561	-2090	0.045	242	1.0	76	46	3400

CFRP was chosen for the pressure bulkheads and honeycomb sandwich panel floor due to its comparably lower density than aluminium, and lack of concern over impact resistance for the internal structure. A bottom-up mass of 3400 kg was calculated from the sized skin, stringers, frames and miscellaneous components. This was within 10% of the initial top-down estimate from Raymer.

13 AERODYNAMICS

For the initial aerodynamic design of Falco the wing was sized using the higher and more accurate MTOW calculated after the PDR using a bottom-up estimation. At first, wing sizing was performed using a constraint diagram (Figure 25), based on the requirements laid out in the specification. Throughout the design iteration it was found that the MTOW and thus wing area were constantly changing. Eventually, to permit design of other aircraft components, the wing area was frozen at 134 m^2 . Whilst appearing large, the high wing area is a direct consequence of the reasons detailed in Section 8.1. This oversized wing area had impacts on the aerodynamic performance, described further in this section.

Several aerofoils' lift-drag characteristics were compared, some of which are shown in Figure 26a. The NASA SC(2) 07xx family was found to have the highest L/D ratio for cruise, with 14% thickness at the root and 10% thickness at the tip. During the PDR, it was made clear that the initial AR of 14 was ambitious for a simple, cantilevered wing and could cause structural issues. In order to get a more accurate AR, a simple rectangular wing model was developed to model how the structural mass of the wing varied as AR was increased. The variations in induced drag and structural mass were shown by plotting Breguet range as a function of aspect ratio, and this found that an AR of 12.25 gave the best trade-off between weight and drag. The combination of the chosen aspect ratio and wing area gave a span of 40.5 m.

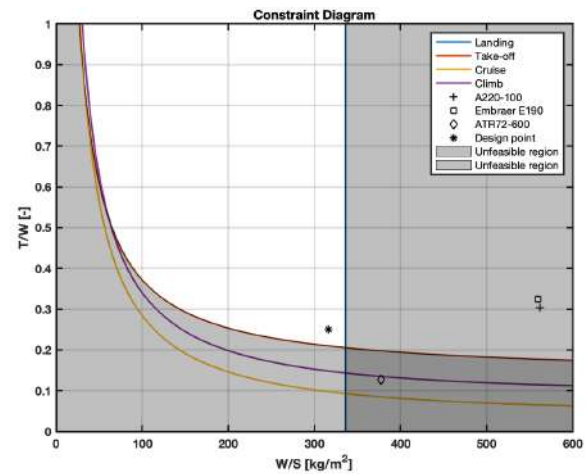
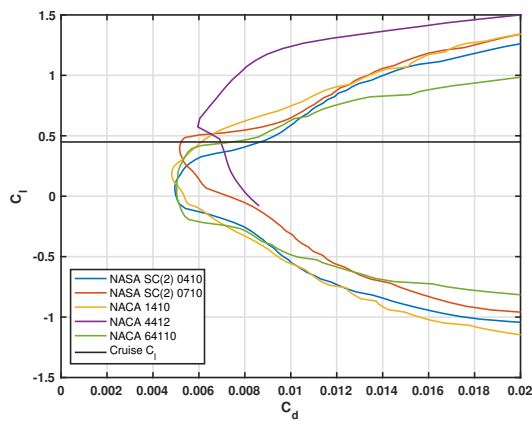
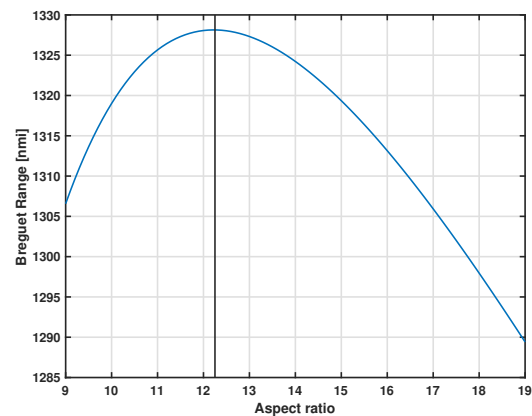


Figure 25: Constraint Diagram



(a) Comparison of Aerofoil Lift-Drag Characteristics

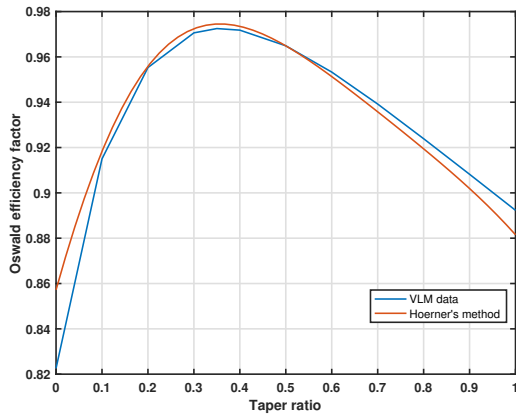


(b) AR vs. Breguet Range Parameter

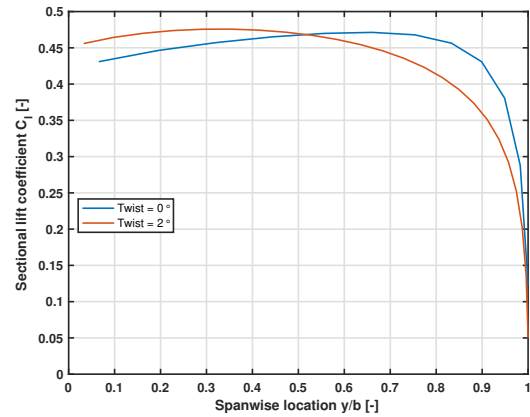
Figure 26

In order to optimise taper ratio, a vortex lattice method was used to evaluate the spanwise lift distribution to give the best Oswald efficiency factor and limit induced drag. As can be seen in Figure 27a, the efficiency peaks at a TR of 0.35, and so this taper ratio was chosen for the Falco wing. The vortex lattice method was validated by using a method from [38], and it can be seen that this method agrees with the vortex lattice method results. The lift distribution of the wing was found to have an initial stall point at 65% span, this meant that control authority would not be maintained in stall.

Therefore, 2° of geometric twist was applied to shift the peak of the lift distribution inboard so that the ailerons would still be effective during a stall.



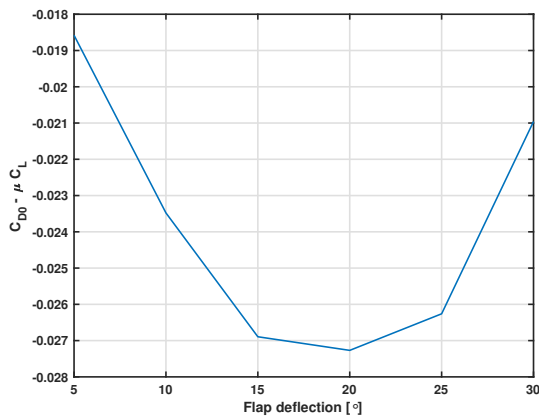
(a) Taper Ratio Optimisation



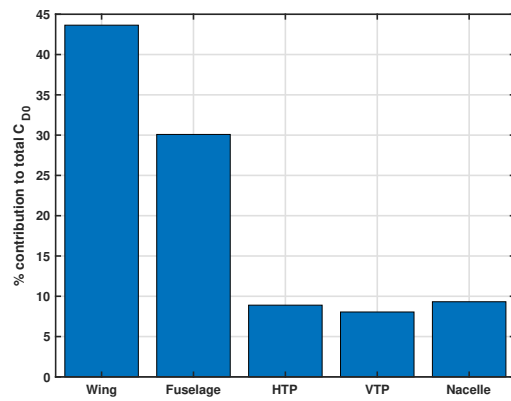
(b) Geometric Twist Optimisation

Figure 27

Having optimised the wing for cruise, attention was put onto the high lift performance of the wing. Given the low stall speed requirement, careful design was necessary to ensure that all the take-off and landing requirements were met. Single slotted flaps and leading-edge slats provide sufficient lift at low speed. Although other turboprop aircraft tend not to utilise leading-edge devices, after some initial calculations it was found that using flaps alone would give an approach speed of 118 kts, which does not meet the requirements. As a result, leading edge slats were included, giving a $C_{L_{max}}$ of 2.54 and an approach speed of 110 kts. In order to find an optimal flap angle for take-off, $C_{D_0} - \mu \cdot C_L$ was plotted as a function of flap deflection angle. As can be seen, this function is minimised at a flap deflection of 20° , and this gives a $C_{L_{TO}}$ of 2.00.



(a) Take-Off Flap Angle Optimisation



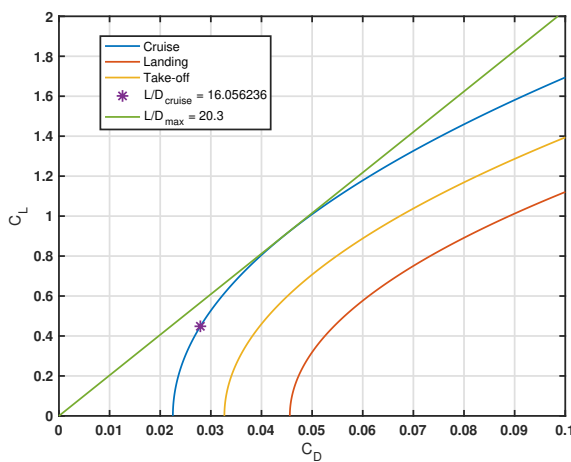
(b) Profile Drag Breakdown

Figure 28

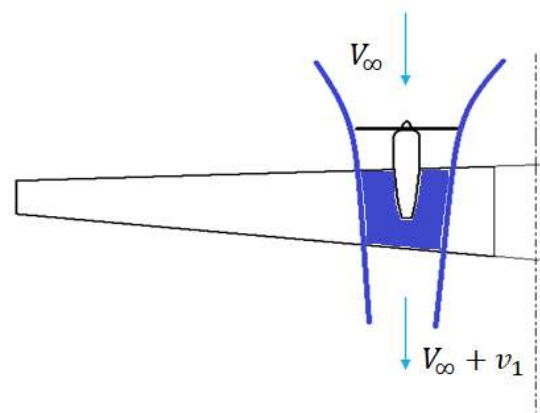
Aerodynamic performance analysis began by calculating an initial value for C_{D_0} , using a drag build-up method described in [29], supplemented with data and methods from ESDU sheets [39]. ESDU sheets were used as it was found that the Raymer method did not account for all types of profile drag, such as the drag due to upsweep at the rear part of the fuselage. A breakdown of the profile drag by component is shown in Figure 28b - it can be seen that the wing and fuselage account for a large amount of the profile drag. The wing mounted nacelles also make a large contribution due to

interference effects between the engine and the wing. Using this drag value, lift-drag polars for the overall aircraft for cruise, take-off and landing could be plotted. This shows how the use of high lift devices greatly increase the profile drag of Falco, and also demonstrates that Falco does not cruise at its optimal aerodynamic efficiency. Figure 29a shows that Falco has a L/D_{max} of 20.3, however L/D at the start of cruise is only 16. This problem is due to the issues stemming from the oversized wing.

The wing mounted turboprop engines influence the lift generated due to an increased dynamic pressure over sections of the wing directly behind the propeller. This effect was investigated using actuator disk theory, where the dynamic pressure was increased by the induced velocity due to the propeller (seen in Figure 29b). It was found that this increase in dynamic pressure resulted in a 5% increase in the overall wing CL at take-off. In future iterations, this effect would be modelled more accurately, and the changes in wing performance reflected in the wing design.



(a) Falco Lift-Drag Polars



(b) Actuator Disk Theory

Figure 29

14 LANDING GEAR

The tricycle configuration was chosen for the landing gear as it is the most stable in the case of landing and taxiing. This configuration is optimal in a trade-off between lateral stability and weight. The NLG will retract forward to save space in a depressurised region. When extended, the CG of the wheel is aft of the strut's CG and thus provides better stability. The MLG is integrated within the wing and when retracted is stowed in the belly fairing, meeting all clearance requirements. An illustration of the stowing kinematics can be seen in Figure 30.

The positions of both the NLG and the MLG relative to the nose can be seen in the GA. Combined with heights of 1.77 m and 2.10 m for the NLG and the MLG respectively, these positions provide the required structural clearances described below. In the event of a NLG collapse the turboprop blades will not contact the ground; there is a clearance of 0.19 m. The LG has a track width of 5.50 m, a main outer wheel span of 6.11 m, and a wheelbase of 12.7 m. These dimensions along with the axle widths of the NLG and the MLG of 0.403 m and 0.607 m, respectively, allow a turn width of 19.9 m. The landing gear design satisfies the ICAO code C requirements (note that the wingspan while taxiing is below 36 m).

The LG height is designed so that in the event of a landing with a high roll angle the wing tip impacts the ground before the blade tips. This is considered the preferred scenario; it is easier to predict the chain of events occurring from the wing tips touching the the ground, as opposed to the blade tip. Deciding the MLG position required close collaboration with the stability engineer, to ensure that the position would remain behind the most aft CG location for the worst case aircraft loading scenarios.

The configuration is designed such that the MLG takes 95% of the load, which enhances both the steering and braking capabilities of the LG. The method from Chai et al. [40] was used to calculate the turnover angle of 46.7° . This method was used over the one given in the reference slides as it allowed reverse engineering calculations, ensuring the position of the LG would also provide sufficient ground stability to meet the turnover angle design requirement of 55° . The tail strike angle was found to be 12.4° , however, it is important to mention that this is the rotation clearance angle, meaning it is the angle when the MLG legs are in the static position. Thus when the MLG struts are extended (before landing and end of take-off), the tip-back angle will be 14.3° . According to Currey [41], most commercial aircraft in the market have tail strike angles below 15° , which is why it was considered as a soft requirement of the design.

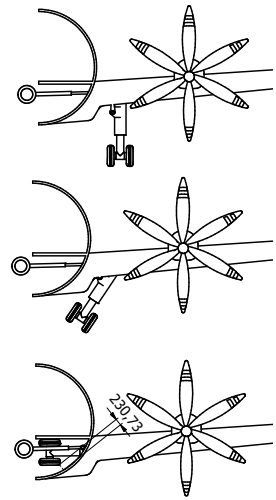


Figure 30: MLG Kinematics

		NLG (Per Tyre)	MLG (Per Tyre)
Max Braking Load	kN	66.7	139
Max Rated Static Load*	kN	29.2	99.1
Max Bottoming Load* (LCX)	kN	17.7 (LC3)	168 (LC3)
Max Horizontal Load* (LCX)	KN	4.75 (LC2)	97.5
Rated Speed*	m/s	50.1	50.1

Table 7: Tyre Loads (* 5% Safety Margin)

The LG tyres were sized using available market tyres in the Goodyear catalogue [42]. The sizes were chosen according to the max. calculated static (rated) and dynamic (bottoming) loads shown in Table 7. The MLG tyres have dimensions of 27.75x8.75 R14.5 in, with two tyres on each MLG strut, or four in total. The NLG dimensions are 26x6.6 R14 in, with two tyres on the strut. Two tyres are used to

satisfy the requirements of the 'one tyre flat' condition. Moreover, the tyres were chosen to be radial over other types for the following reasons [41]:

- They offer higher flotation
- Have 40-60% tread wear improvement
- Offer weight reduction of 12-28% to equivalent bias-ply tyres, which leads to cost savings

The ACN was calculated using the 'COMFAA 3.0' software by editing the MLG axle width to be 0.64 m, along with other parameters such as gross weight and tyre pressure. It was found to be 25.1 for flexible ICAO Code C. This value was proven to be in line with other aircraft of similar MTOWs according to [43]. The BAC-111 Series 400, which has a slightly higher MTOW than Falco, has an ACN of 27 for Code C. While the BAe-146-100, with a slightly lower MTOW than Falco, has an ACN of 23 for ICAO code C.

The static rated loads were calculated using methods from Currey [41]. While the bottoming loads were determined based on calculations of the 3 suggested load cases; 1- CS 25.493, 2- CS 25.495, 3- CS 25.491 [31]. It was determined that CS 25.495 (0.5g turning case) was the load case that subjected both the MLG tyres and the NLG tyres to the highest bottoming loads, shown in Table 7. Note the given rated speed is the speed the tyres must withstand and is not the same as the rated speed from the Goodyear catalogue [42].

The brakes of the MLG were sized using the provided MATLAB sizing code 'LGbrakesizing.m'. Carbon brakes were chosen over equivalent steel brakes for mass reduction purposes. The 14.5 in (0.368 m) rims will have a width equal to that of the wheel (8.75 in, 0.222 m), this would give them the ability to house the brakes, which require a width of 0.153 m and an outer diameter of 0.305 m.

The LG shock absorber (SA) allows it to withstand all the harsh landing conditions the structure is subjected to. A sink rate of 10 ft/s (3.048 m/s) was used to calculate the stroke length (X_s) using Equation 1. X_t is the tyre deflection (0.403 m); the dynamic reaction factor (λ) was assumed to be 1 for the size of our aircraft. An X_s of 0.362 m was calculated, the rest of the shock absorber dimensions were derived from these numbers and are all given in Table 9. It is important to mention that the value of X_s is low, due to Falco having a smaller MTOW compared to other commercial aircraft. A gas-spring curve of this oleo-pneumatic tube was plotted (Figure 31) to confirm primarily that the pressure inside the tube will not exceed 6000 psi (especially under dynamic loads), thus proving the SA will not break or start leaking. Secondly, it demonstrates that the static pressure chosen would be neither too high nor too low, such that the LG would not 'stick to' nor 'bounce off' the runway during landing.

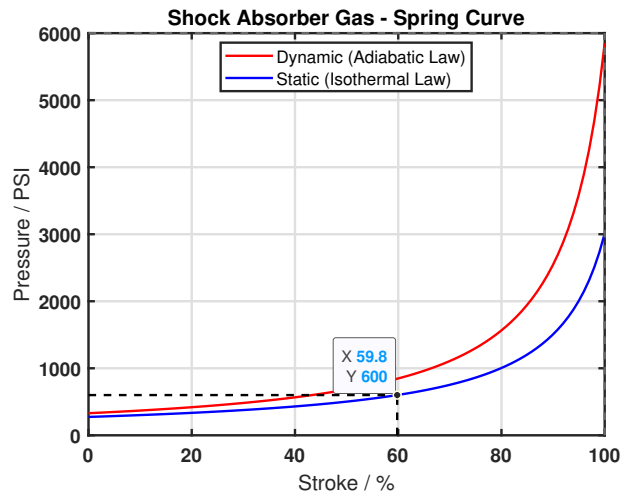


Figure 31: Gas Spring Curve Oleo-Pneumatic SA

$$X_s = \frac{\frac{V^2}{2\lambda g} - 0.5 \cdot X_t}{0.75} \quad (1)$$

The mass of the LG structure was calculated using its volume and the density of the material used in each structure. All the LG mass details could be found in Table 8. It is important to address that Falco's LG is responsible for a higher percentage of MTOW than the usual 5%. This is because the

	[kg]
Main LG	1701
Nose LG	733
Total	2434
% MTOW	~6.13 %
MLG % of MTOW:	4.28
NLG % of MTOW:	1.85

Table 8: LG Mass Breakdown

Shock Absorber Dimension		Values
Stroke Length	m	0.362
Reserve Length (10% of X_s)	m	0.0362
Static Pressure	PSI	600
Internal diameter	m	0.241
Area of sliding tube	m ²	0.0454
Compressed Volume	m ³	0.00165
Extended Volume	m ³	0.0181
Adiabatic Constant (Dynamic)	-	6100
Isothermal Constant (Static)	-	13,600
Calculated Force on one MLG leg	kN	188
Max Static Pressure	PSI	3010
Max Dynamic Pressure	PSI	5850

Table 9: Shock Absorber Data

MLG struts had to be made longer to ensure they reach the belly fairing while maintaining a good enough main outer wheel span to satisfy the turnover angle requirement.

Due to the nature of the loads the LG will be subjected to, plus the fact they are a single load path structure, it was crucial for the team to choose a material with high strength to be able to absorb such loads. Thus, titanium was chosen for all the main structural components such as the rims and SA. To aid mass reduction, the components that do not take load from the main path are made from alternative materials. For example, carbon and composites for the brakes and support braces respectively. AerMet 100 and 4043 steels were considered instead of Titanium as they are 6x cheaper and would therefore align with our DOC design driver [44,45]. However, a comparison showed that the use of steel would increase the weight of the LG to 10.5% of MTOW, which shifts the CG into an unacceptable region. [46,47].

Finally to analyse the stresses in the LG, preliminary Euler buckling was performed using methods found in [48]. The analyses were conducted for both titanium alloy and AerMet 100 steel. Both metals proved to have greater yield strengths than the stresses the LG will be subjected to. The titanium alloy components had a maximum allowable load of 4.06×10^6 kN and 8.55×10^5 kN for the MLG and the NLG structures respectively.

15 PROPULSION AND PERFORMANCE

The major design decision between the choice of turboprop or turbofan engines was dominated by fuel burn. To find this, an analytical fuel burn script was used, which was based on inverse Breguet-range fuel burn calculations. The script assumes the aircraft is flying in a trimmed condition, where the total drag is a function of the zero-lift drag and the induced-drag from the main wing. Due to the lack to high fidelity aerodynamic simulation, there is no drag rises included. Hence, the script ignores the effect of drag from the tail plane and propeller at varying speeds. It must be noted, that values for fuel burn during taxi, take-off, approach and landing were provided in the specification.

Ignoring mass and aerodynamic differences between the two engines, the turboprop resulted in a 13% reduction in fuel burn over the design mission, Figure 5. In conjunction with improved climb performance, this was significant enough to choose the turboprop as the powerplant for Falco. Engine scaling was governed by the BFL requirements, in conjunction with an aim for low fuel burn. Results from the described method showed that a smaller engine resulted in a lower fuel burn at the cost of SFC efficiency, hence the engine was scaled to meet the BFL requirement of 1500 m.

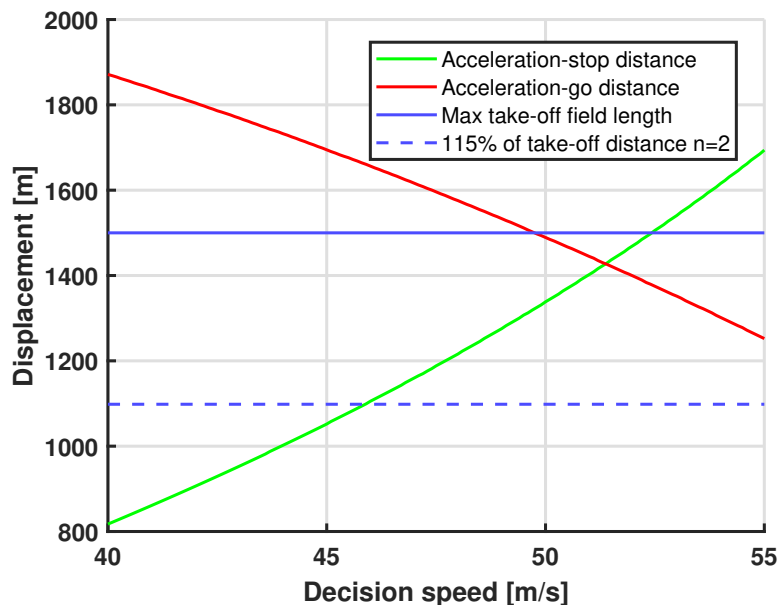


Figure 32: Balanced Field Length

To find the BFL, an discrete time-stepping method was devised for most accurate results, found to be within 3% of actual aircraft data. The engine features a variable pitch propeller, which is assumed to be optimally trimmed at each condition. In conjunction, performance margins for a 2% increase in MTOW, a 1.5% reduction in aerodynamic parameters and a 1.5% increase in SFC were used. An engine scaling of 0.9 was found to meet this requirement. In the initial weeks, a scaling of 0.85 was chosen using lower fidelity BFL equations, without accounting for further regulations of 2.4% and 1.2% climb gradients. Falco has a resulting BFL of 1430m, with a max climb gradient of 3%, exceeding both the 2.4% and 1.2% regulations, all of which is >115% of the aircraft stall speed.

The aircraft is designed to fly at an initial and final cruise altitude of 25,000 ft which is reached in 14 minutes at 200 kts CAS, cruising below 26,000 ft to mitigate contrail formation and reduce fuel burn by operating close to max L/D. The aircraft cruises at an L/D of 16, below the max L/D of 20.3. This is a result of freezing the wing and engine scaling early on, to allow for detailed design of other components. To overcome the non-optimal cruise conditions, the aircraft could fly at a greater altitude, which risks contrail formation. Alternatively, a lower cruise Mach could be used, but this

would increase block time and reduce the surge margin that the aircraft is operating at.

Contrail formation is not fully understood and exact prediction methods are not available; research conducted by NASA in 2002 revealed that the Appleman's chart is 98% accurate at predicting when formation will not occur [49]. Given European atmospheric condition charts, this can be seen to be as low as 26,000 ft. It must be noted that Appleman's prediction is a function of pressure and temperature, however in reality, it is the result of many more variables, such as by-pass ratio. Air traffic instructions could also pose restrictions on the aircrafts altitude and direction, which could alter the mitigation.

At cruise Falco travels at Mach 0.47, optimised between block fuel and block time, whilst allowing the engine to operate with a sufficient surge margin. As a result, for the 800 nm design mission Falco has an SFC at mid cruise of 0.0413 and a total fuel mass of 4250 kg at MTOW. The breakdown can be seen in Figure 33a. The full mission profile is shown in Figure 33b, compared with the current generation ATR72-600.

Using equations based on historical data of a wide range of aircraft, a noise estimation was conducted. Engine thrust, propeller size and flap area were a few of the main variables. The results of this can be seen in Table 2.

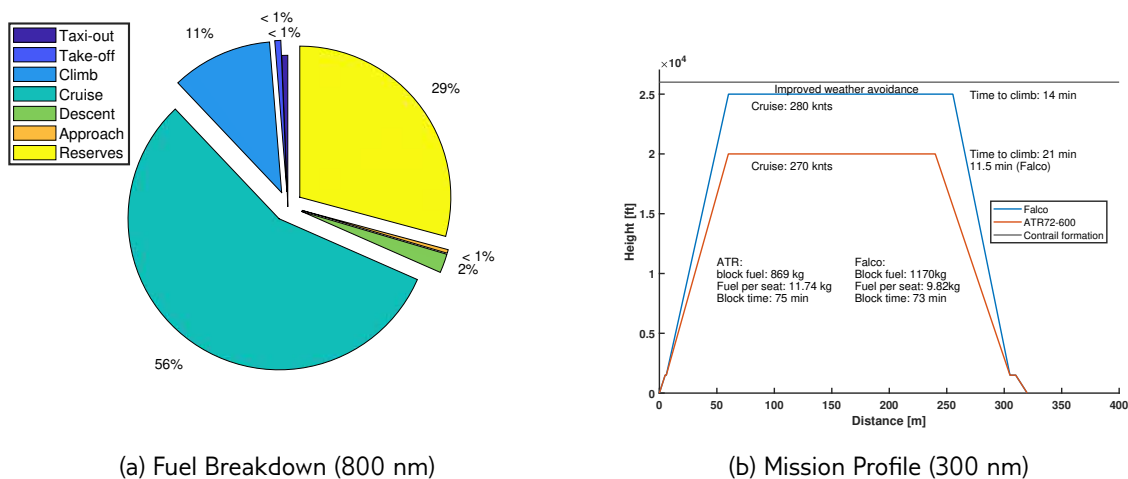


Figure 33

Further, in the event of engine failure, Falco has a BFL of 1430 m, with a decision speed of 51.3 m/s (87% of take-off speed), which can be seen in Figure 32. This can be maintained to a one engine inoperative altitude of 16,000 ft, which allows for sufficient clearance when flying over the northern Italian Alps. The aircraft also meets the landing requirements with a JAR landing field of 1074 m, allowing for 50 ft object clearance and a 3-degree descent slope. This was calculated using an analytical discrete time-stepping method, found to be within 2% of actual aircraft data for a braking co-efficient of $\mu_b = 0.25$. This methods ignores the effect of spoilers due to modelling difficulties. Due to the low approach speed, the landing field length was not found to be restrictive or a design factor in the performance.

The key design drivers for the fuel system were: available fuel tank volume, wing bending relief and integration with other systems in the wing box. From the mission profile including the reserve requirements, the tanks were designed to carry 4250 kg or 5315 L of fuel, using a standard aviation fuel density of 0.8 kg/L. Falco has a total of six tanks, two main tanks and one surge tank per wing. This is to avoid fuel spillage and allow air to exit the tanks during refuelling. As shown in Figure 34, the inner tank is placed between the 5th and 12th rib, the outer tank between the 13th and 17th rib

and the surge tank between the 18th and 19th rib. The wing ribs help reduce sloshing which could promote undesirable dynamic effects during manoeuvres and gusts. The placement of the tanks depended on the landing gear storage and rotor disk burst regions. Figure 34 shows that in the case of outboard turbine burst, the two main tanks and the rear spar are entirely outside the 5° region. For inboard bursts, the tanks on the opposite wing may be affected, however the tanks outboard of the failed engine will remain unaffected. The cross-feed lines and valves were designed to make sure that each tank can independently deliver fuel to both engines. The rear spar on both wings will also remain unaffected. The inner tanks are reinforced as the hot exhaust gases from the engine flow over that section of the wing. Falco is fitted with an on-board inert gas generating system (OBIGGS) to make the fuel tanks safer [50].

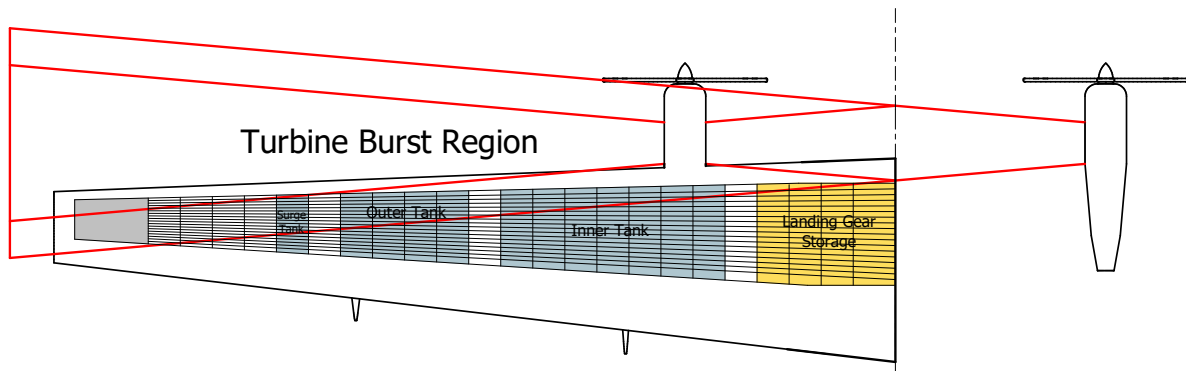


Figure 34: Falco Wing Box, LG and Fuel Systems Integration

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